

#1 · Predictive Analytics Problem Definition

Descriptive analytics

Answer

Analysis that answers what happened, using summaries and aggregations of historical data without fitting a model or forecasting a future outcome.

#2 · Predictive Analytics Problem Definition

Predictive analytics

Answer

Analysis that answers what is likely to happen, using patterns learned from historical data to forecast an outcome that is not yet known.

#3 · Predictive Analytics Problem Definition

Prescriptive analytics

Answer

Analysis that answers what action should be taken, using a model's predictions as an input to a decision rule or optimization.

#4 · Predictive Analytics Problem Definition

Why does prescriptive analytics depend on predictive analytics?

Answer

A decision cannot be prescribed from a prediction that has not yet been produced, so prescription is always downstream of prediction.

#5 · Predictive Analytics Problem Definition

Why review data descriptively before modeling it?

Answer

A descriptive review can catch data errors, such as an inflated claim count from a data entry mistake, before those errors corrupt a fitted model.

#6 · Predictive Analytics Problem Definition

Target variable

Answer

The outcome a predictive model is trying to estimate, such as claim severity, lapse status, or claim count.

#7 · Predictive Analytics Problem Definition

Continuous target variable

Answer

A numeric outcome that can take any value in a range, such as the ultimate loss amount on a property claim.

#8 · Predictive Analytics Problem Definition

Binary target variable

Answer

An outcome with exactly two possible categories, such as whether a policy lapses within twelve months.

#9 · Predictive Analytics Problem Definition

Count target variable

Answer

An outcome measured as a nonnegative whole number, such as the number of claims a policyholder files in a year.

#10 · Predictive Analytics Problem Definition

Why not fit ordinary linear regression to a binary target?

Answer

Linear regression can produce fitted values outside zero and one, which violates the meaning of a probability, so logistic regression or a classification tree is used instead.

#11 · Predictive Analytics Problem Definition

Supervised learning

Answer

A modeling problem where the training data includes the actual historical value of the target variable, allowing predictions to be compared against known outcomes.

#12 · Predictive Analytics Problem Definition

Unsupervised learning

Answer

A modeling problem with no labeled target variable, where the goal is to find structure among the predictor variables themselves, such as clustering.

#13 · Predictive Analytics Problem Definition

Unit of observation

Answer

What a single row of the data set represents, such as a policy-year, an individual vehicle, or a claimant, and choosing it wrong can silently bias a model.

#14 · Predictive Analytics Problem Definition

Risk of an incorrect unit of observation

Answer

Treating correlated observations, such as three vehicles in one household, as independent can understate the true uncertainty in a model's estimates.

#15 · Predictive Analytics Problem Definition

Static snapshot problem

Answer

A predictive problem, such as underwriting approval at application, that uses a single point-in-time view of the data rather than a time series.

#16 · Predictive Analytics Problem Definition

Longitudinal modeling problem

Answer

A predictive problem involving a time dimension, such as estimating a workers compensation claim's ultimate cost at multiple development periods.

#17 · Predictive Analytics Problem Definition

Why might missing data itself carry information?

Answer

A missing response, such as an unanswered smoking status question, may be more common among applicants for whom the true answer is unfavorable, so the missingness pattern is informative.

#18 · Predictive Analytics Problem Definition

Bias in a predictive model

Answer

Systematic error that arises when a model's functional form is too simple to represent the true relationship between predictors and target.

#19 · Predictive Analytics Problem Definition

Variance in a predictive model

Answer

Error that arises when a model is so flexible that it fits the noise in one particular training sample, producing very different results across different samples.

#20 · Predictive Analytics Problem Definition

Bias-variance tradeoff

Answer

As model complexity increases, bias tends to fall while variance tends to rise, so total error is minimized at an intermediate complexity level rather than at either extreme.

#21 · Predictive Analytics Problem Definition

Irreducible error

Answer

The portion of prediction error caused by inherent randomness in the outcome itself, which no model, however sophisticated, can eliminate.

#22 · Predictive Analytics Problem Definition

Expected total prediction error decomposition

Answer

Expected prediction error equals squared bias plus variance plus irreducible error.

#23 · Predictive Analytics Problem Definition

Example of high bias in an actuarial model

Answer

Fitting a straight line to model auto claim frequency against driver age when the true relationship is U-shaped, so the line cannot capture the pattern regardless of sample size.

#24 · Predictive Analytics Problem Definition

Example of high variance in an actuarial model

Answer

A decision tree grown to full depth with only two or three claims per leaf, which fits training noise and would look very different if retrained on a new random sample.

#25 · Predictive Analytics Problem Definition

Does adding more training data fix high bias?

Answer

No, because high bias comes from the model's structure being too rigid, not from insufficient data.

#26 · Predictive Analytics Problem Definition

Overfitting

Answer

A model has learned the noise specific to its training sample rather than the generalizable signal, shown by low training error but high validation error.

#27 · Predictive Analytics Problem Definition

Underfitting

Answer

A model is too simple to capture the true pattern, shown by training error and validation error that are both high and similar to each other.

#28 · Predictive Analytics Problem Definition

Training error as complexity increases

Answer

Training error decreases monotonically as model complexity increases, since a more flexible model can always fit the training data at least as well as a simpler one.

#29 · Predictive Analytics Problem Definition

Validation error as complexity increases

Answer

Validation error typically falls, reaches a minimum, then rises again as complexity keeps increasing past the point where variance begins to dominate bias.

#30 · Predictive Analytics Problem Definition

Signature of overfitting on training versus validation error

Answer

A large gap where training error is low but validation error is high indicates the model has overfit the training sample.

#31 · Predictive Analytics Problem Definition

Signature of underfitting on training versus validation error

Answer

Training error and validation error that are both high and close to each other indicate the model is underfit.

#32 · Predictive Analytics Problem Definition

Example of underfitting in mortality modeling

Answer

Using only attained age as a predictor when smoking status and build also carry mortality signal produces systematically biased premium rates.

#33 · Predictive Analytics Problem Definition

Example of overfitting in mortality modeling

Answer

Including a three-way interaction of age, zip code, and month of issue that fits quirks in past data but has no genuine mortality basis and will not generalize.

#34 · Predictive Analytics Problem Definition

Model complexity

Answer

The flexibility of a model, driven by its number of parameters or splits, that determines how closely it can bend its fit to the training data.

#35 · Predictive Analytics Problem Definition

Why translate a vague business question into a statistical question?

Answer

A vague request rarely specifies a target variable, population, time period, or measurable success criterion, none of which a model can answer without being made explicit.

#36 · Predictive Analytics Problem Definition

Example vague business question

Answer

"Our homeowners book is losing money, figure out why" does not specify a target variable, population, or definition of losing money.

#37 · Predictive Analytics Problem Definition

What must a good statistical translation make explicit?

Answer

The target variable, the population being studied, the time window, and how the model's output will actually be used.

#38 · Predictive Analytics Problem Definition

Data availability constraint on a model's information

Answer

A model cannot use information that will not exist yet at the moment the prediction actually needs to be made.

#39 · Predictive Analytics Problem Definition

Example of a leakage-style timing error

Answer

Using next year's claim experience to predict this year's lapse would use information not available at prediction time and is invalid.

#40 · Predictive Analytics Problem Definition

Why exclude policies still inside a surrender charge period from a lapse model?

Answer

Lapse dynamics differ sharply for policies inside a surrender charge window, so mixing them with post-surrender policies would blur the population being modeled.

#41 · Predictive Analytics Problem Definition

Lead time in a lapse model deployment

Answer

A lapse model should be scored far enough ahead of the likely lapse date, such as three to six months, to give a retention program time to act.

#42 · Predictive Analytics Problem Definition

Factor: available data and technology

Answer

A statistical question is only pursuable if the data it needs actually exists, at an adequate grain and history, and the company's systems can process it.

#43 · Predictive Analytics Problem Definition

Example of a data availability limitation

Answer

A model that ideally uses telematics hard-braking data cannot be built if the insurer has not yet rolled out a telematics program.

#44 · Predictive Analytics Problem Definition

Factor: significance of business impact

Answer

A modeling project's expected dollar impact should be weighed against its cost in time and disruption before it is prioritized over other candidate projects.

#45 · Predictive Analytics Problem Definition

Why prioritize a large premium line with an adverse loss ratio over a small line?

Answer

The potential improvement in decisions on a large, underperforming line typically produces far greater dollar impact than the same modeling effort on a minor line.

#46 · Predictive Analytics Problem Definition

Factor: implementation challenges

Answer

A model that cannot actually be deployed into production systems, filed with regulators, or maintained over time provides little practical value regardless of its accuracy.

#47 · Predictive Analytics Problem Definition

Why are GLM coefficients easier to implement than a random forest in a legacy rating engine?

Answer

GLM coefficients translate naturally into rating factors a legacy system can apply, while a forest of thousands of trees is difficult or impossible to embed the same way.

#48 · Predictive Analytics Problem Definition

Why does regulatory filing favor interpretable models?

Answer

Many US insurance regulators require a rating algorithm to be explained and reviewed, which favors models whose logic can be stated in plain terms over opaque, highly complex models.

#49 · Predictive Analytics Problem Definition

Model maintenance as an implementation concern

Answer

A model that must be refit regularly as new experience emerges needs a sustainable production pipeline, not just a single one-time analysis.

#50 · Predictive Analytics Problem Definition

Next step when additional predictor data is unavailable

Answer

Identify what data, such as credit-based insurance scores or telematics mileage, would most improve the analysis and recommend the company pursue obtaining it.

#51 · Predictive Analytics Problem Definition

Why disclose a model's limitations rather than presenting it as final?

Answer

A model trained during one economic environment, such as low interest rates, may not generalize once conditions change, so its limits should be stated honestly.

#52 · Predictive Analytics Problem Definition

Appropriate response to systematic underprediction in one segment

Answer

Investigate whether an important segment-specific predictor is missing, such as a more granular occupation code, rather than discarding the model outright.

#53 · Predictive Analytics Problem Definition

Recognizing a valuable follow-up business question

Answer

An analysis can reveal a narrower, more actionable follow-up project, such as a roof-age-triggered inspection program discovered while modeling claim frequency.

#54 · Predictive Analytics Problem Definition

Role of problem definition in an analytics project

Answer

Problem definition determines whether every later step is aimed at a question worth answering that can actually be answered with the data and technology on hand.

#55 · Predictive Analytics Problem Definition

Why does the real Exam PA emphasize written judgment over calculation?

Answer

The exam is graded on how well a candidate reasons through a business scenario and communicates a recommendation, not on producing a single numeric answer.

#56 · Predictive Analytics Problem Definition

Categorical target with more than two levels

Answer

A target variable with more than two categories, such as which of five injury severity categories a workers compensation claim falls into; note such categories are often ordinal (naturally ordered by severity) rather than purely unordered, which can affect model choice.

#57 · Predictive Analytics Problem Definition

Why can descriptive analytics prevent wasted modeling effort?

Answer

If the underlying business request only needs an explanation of what already happened, building a complex predictive model is unnecessary over-engineering.

#58 · Predictive Analytics Problem Definition

Clustering

Answer

An unsupervised technique that groups observations, such as policyholders, into segments based on similarity in their predictor variables, with no predefined correct label.

#59 · Predictive Analytics Problem Definition

Why must the unit of observation match the true risk structure?

Answer

If the unit of observation is finer than the true independent risk unit, such as vehicle-year instead of household, correlated observations can be mistaken for independent ones.

#60 · Predictive Analytics Problem Definition

Why choose a modeling technique after identifying target variable type?

Answer

The mathematical form of the target variable, continuous, binary, or count, constrains which techniques are valid, so identifying it comes before choosing a method.

#61 · Predictive Analytics Problem Definition

One model per horizon versus a single model with time as a predictor

Answer

A longitudinal claim development problem can be handled either by fitting a separate model at each development period or by including development time as a predictor in one model.

#62 · Predictive Analytics Problem Definition

Practical effect of high bias on rates

Answer

A biased model produces systematically wrong estimates in a consistent direction across many possible training samples, such as consistently underpriced rates for a mispriced segment.

#63 · Predictive Analytics Problem Definition

Practical effect of high variance on model stability

Answer

A high-variance model would produce meaningfully different predictions if it were refit on a different random sample of the same underlying population.

#64 · Predictive Analytics Problem Definition

Sweet spot of model complexity

Answer

The intermediate complexity level at which the sum of squared bias and variance, and therefore total expected error, is minimized.

#65 · Predictive Analytics Problem Definition

How does pruning a decision tree affect bias and variance?

Answer

Pruning reduces model complexity, which tends to increase bias slightly but reduce variance, often lowering total validation error if the tree had been overfit.

#66 · Predictive Analytics Problem Definition

How does adding a relevant predictor typically affect bias?

Answer

Adding a genuinely relevant predictor that was previously omitted tends to reduce bias by letting the model represent the true relationship more accurately.

#67 · Predictive Analytics Problem Definition

Risk of adding an irrelevant predictor

Answer

Adding a predictor with no true relationship to the target tends to increase variance without reducing bias, worsening the bias-variance tradeoff.

#68 · Predictive Analytics Problem Definition

Why must an actuary state assumptions explicitly in a written recommendation?

Answer

A business stakeholder reading the memo needs to see exactly what question was answered, which may be narrower than the question they originally asked.

#69 · Predictive Analytics Problem Definition

Interpretability as a business requirement, not just a preference

Answer

Interpretability is often a hard implementation constraint, since regulators, business users, and legacy systems may require a model whose logic can be explained plainly.

#70 · Predictive Analytics Problem Definition

Example translation of a marketing request into a modeling question

Answer

"Reduce lapses" can be translated into predicting a binary lapse indicator over the next policy year for in-force policies past the surrender charge period, scored months in advance.

#71 · Predictive Analytics Problem Definition

Why weigh cost of staff time and disruption against expected impact?

Answer

A technically feasible project is not automatically worth pursuing if its expected business benefit is small relative to the actuarial time and operational change it requires.

#72 · Predictive Analytics Problem Definition

When is a model considered mature versus a permanent finished artifact?

Answer

A model is mature when it is monitored and periodically refit as new experience emerges, rather than treated as a static, never-revisited output.

#73 · Predictive Analytics Problem Definition

Why does grain of historical data matter for feasibility?

Answer

A claims system that only retains detailed data for a limited history restricts how far back a credible training data set can reach.

#74 · Predictive Analytics Problem Definition

Relationship between problem definition and validation results

Answer

Validation results, such as a segment with systematic underprediction, can reveal that the original problem definition omitted an important predictor and needs revisiting.

#75 · Predictive Analytics Problem Definition

What three questions distinguish descriptive, predictive, and prescriptive analytics?

Answer

Descriptive asks what happened, predictive asks what is likely to happen, and prescriptive asks what action should be taken.

#76 · Data Exploration and Visualization

Structured data

Answer

Data organized into a fixed set of fields, usually rows and columns, where every record has the same layout, such as a personal auto policy file with columns for driver age, vehicle type, and territory.

#77 · Data Exploration and Visualization

Unstructured data

Answer

Data with no fixed schema, such as adjuster notes, call transcripts, inspection images, or free-text accident descriptions, which cannot be fed directly into a model until converted into structured features.

#78 · Data Exploration and Visualization

Semi-structured data

Answer

Data that has some organization but not the strict tabular shape of a spreadsheet, such as a JSON file with nested claim and coverage objects.

#79 · Data Exploration and Visualization

Why can't adjuster notes be used directly in a GLM?

Answer

They are unstructured text with no fixed schema; they must first be converted into structured features, such as keyword counts, phrase flags, or a sentiment score.

#80 · Data Exploration and Visualization

Give an example of a mixed structured/unstructured Exam PA data set.

Answer

A workers' compensation study might provide a clean claims table (structured) alongside a text field of injury descriptions (unstructured).

#81 · Data Exploration and Visualization

Nominal variable

Answer

A categorical variable with no natural order, such as vehicle color, marital status, or state of residence.

#82 · Data Exploration and Visualization

Ordinal variable

Answer

A categorical variable with a natural order but no consistent numeric spacing between levels, such as a satisfaction rating of low, medium, or high.

#83 · Data Exploration and Visualization

Interval variable

Answer

A variable with consistent numeric spacing but no true zero point, so ratios are not meaningful, such as calendar year.

#84 · Data Exploration and Visualization

Ratio variable

Answer

A variable with a true zero point and meaningful ratios, such as claim amount, annual premium, or driver age.

#85 · Data Exploration and Visualization

Why is calendar year an interval variable rather than a ratio variable?

Answer

It has consistent spacing between years but no true zero, so year 2000 is not meaningfully twice year 1000.

#86 · Data Exploration and Visualization

Why is claim amount a ratio variable?

Answer

It has a true zero (no claim occurred) and ratios are meaningful, so a \$2,000 claim really is twice as large as a \$1,000 claim.

#87 · Data Exploration and Visualization

Response variable

Answer

The outcome being predicted in a modeling exercise, also called the target or dependent variable, such as whether a policy lapses or the size of a claim.

#88 · Data Exploration and Visualization

Predictor variable

Answer

An input used to predict the response, also called an explanatory variable, independent variable, or feature, such as policyholder age or territory.

#89 · Data Exploration and Visualization

Continuous variable

Answer

A variable that can take any value within a range, such as loss amount or driver age measured in exact years.

#90 · Data Exploration and Visualization

Discrete variable

Answer

A variable that only takes distinct, separate values, such as number of claims or number of dependents.

#91 · Data Exploration and Visualization

What is the difference between continuous/discrete and nominal-through-ratio classifications?

Answer

Continuous and discrete describe how a variable is measured, while nominal, ordinal, interval, and ratio describe what mathematical operations are meaningful on it.

#92 · Data Exploration and Visualization

Give a complete classification of claim count.

Answer

Claim count is a discrete ratio variable: it takes distinct values and has a true zero with meaningful ratios.

#93 · Data Exploration and Visualization

What three design questions should an analyst evaluate before exploring individual variables?

Answer

Whether the data's time frame, sampling method, and granularity are appropriate for the modeling purpose.

#94 · Data Exploration and Visualization

Time frame (data design)

Answer

Whether the period the data covers is still representative of the risk being modeled today, since stale data can look statistically strong while reflecting outdated conditions.

#95 · Data Exploration and Visualization

Why might a homeowners data set from five to fifteen years ago be problematic?

Answer

It may not reflect current construction costs, current weather patterns, or current underwriting rules, so a model built on it could be practically wrong even if statistically strong.

#96 · Data Exploration and Visualization

Why might a data time frame that is too short be a problem?

Answer

It may fail to capture rare but important events, such as a hurricane season or a pandemic-driven change in claim frequency.

#97 · Data Exploration and Visualization

Sampling (data design)

Answer

Whether the data set includes every eligible record or only a subset, and if a subset, whether that subset is representative of the full population.

#98 · Data Exploration and Visualization

Selection bias (in data sampling)

Answer

Bias introduced when a data set only contains a non-representative slice of records, such as only disputed or litigated claims, skewing a severity model toward extreme claims.

#99 · Data Exploration and Visualization

Give an example of selection bias from a claims data set.

Answer

If a company only retained disputed or litigated claims, a severity model built from that file would be badly biased toward extreme claims, since ordinary quickly settled claims never entered the sample.

#100 · Data Exploration and Visualization

Granularity (data design)

Answer

The level of detail of each row in a data set, such as whether a row represents a single transaction, a claim, a policy, or an aggregated month of experience.

#101 · Data Exploration and Visualization

Give an example of a granularity mismatch.

Answer

A response variable defined at the policy level (did the policyholder lapse) but a predictor only available at a more aggregated territory level.

#102 · Data Exploration and Visualization

How is a granularity mismatch commonly resolved?

Answer

By aggregating the more detailed data, such as claim-level records, up to the coarser level, such as policy level, before merging it with other characteristics.

#103 · Data Exploration and Visualization

Which textbook's ideas underlie Exam PA's grading rubrics for exploratory graphs?

Answer

Kieran Healy's 'Data Visualization: A Practical Introduction.'

#104 · Data Exploration and Visualization

Proportional ink principle

Answer

The idea, associated with Edward Tufte, that the amount of ink used to represent a data value should be proportional to the value itself.

#105 · Data Exploration and Visualization

Why does starting a bar chart's y-axis at a nonzero value violate the proportional ink principle?

Answer

It stretches a small true difference between bars into what visually looks like a huge difference, misleading the reader.

#106 · Data Exploration and Visualization

Give the chapter's example of a proportional-ink violation.

Answer

Comparing average claim severity of \$4,200 versus \$4,350 by starting the y-axis at \$4,000 instead of \$0 makes an actual 3.5% difference look enormous.

#107 · Data Exploration and Visualization

Chartjunk

Answer

Any visual element that adds decoration without adding information, such as unnecessary 3D effects, heavy gridlines, redundant legends, or ornamental images.

#108 · Data Exploration and Visualization

Why is chartjunk harmful beyond wasting space?

Answer

It actively competes with the reader's attention against the data itself, making the actual pattern harder to see.

#109 · Data Exploration and Visualization

Which chart type suits a single continuous variable?

Answer

A histogram or a boxplot, such as for exploring claim severity.

#110 · Data Exploration and Visualization

Which chart type suits a single categorical variable?

Answer

A bar chart, such as for the distribution of policies by vehicle class.

#111 · Data Exploration and Visualization

Which chart type suits two continuous variables?

Answer

A scatterplot, such as age plotted against annual premium.

#112 · Data Exploration and Visualization

Which chart type suits two categorical variables?

Answer

A grouped or faceted bar chart, or a mosaic plot, such as coverage type against claim outcome.

#113 · Data Exploration and Visualization

Name two commonly exam-flagged chart-type errors.

Answer

Using a pie chart to compare many categories, and using a line chart to connect points that have no real ordering.

#114 · Data Exploration and Visualization

Name three ways a graph's axes can mislead beyond a truncated y-axis.

Answer

Using inconsistent bin widths on a histogram, reversing an axis direction without labeling it, or using a log scale without telling the reader.

#115 · Data Exploration and Visualization

What is Healy's broader point about the purpose of a graph?

Answer

A graph is a communication tool aimed at a specific audience, not a decoration, and its complexity should match that audience.

#116 · Data Exploration and Visualization

How should chart complexity differ between an actuarial peer and a business stakeholder audience?

Answer

A graph for an actuarial peer reviewing model diagnostics can carry more technical density than one built for a non-technical stakeholder deciding on a rating plan.

#117 · Data Exploration and Visualization

Univariate exploration

Answer

Examining one variable at a time, almost always the first concrete step in an Exam PA written response.

#118 · Data Exploration and Visualization

Histogram

Answer

A graph that groups a continuous variable into bins and shows the count or proportion of observations in each bin.

#119 · Data Exploration and Visualization

What shape does a claim severity histogram typically show?

Answer

A right-skewed shape, where most claims are modest in size but a long thin tail of very large claims stretches to the right.

#120 · Data Exploration and Visualization

Why does a right-skewed severity histogram matter for model choice?

Answer

It signals that a normal distribution assumption would fit poorly, pointing instead toward a gamma or lognormal model.

#121 · Data Exploration and Visualization

Boxplot

Answer

A summary of a continuous variable using five numbers: minimum (excluding outliers), first quartile, median, third quartile, and maximum (excluding outliers), with outliers plotted as individual points.

#122 · Data Exploration and Visualization

What is a boxplot especially useful for?

Answer

Comparing the same variable across several groups side by side, such as claim severity across five vehicle classes.

#123 · Data Exploration and Visualization

What does a longer whisker and more outlier points for sports cars versus minivans suggest?

Answer

A quick visual confirmation that sports cars have more variable and extreme claim severity, consistent with underwriting expectations.

#124 · Data Exploration and Visualization

Why is the median often more representative than the mean for skewed severity data?

Answer

The mean is pulled upward by a long right tail of large claims, while the median better reflects the typical claim.

#125 · Data Exploration and Visualization

Why can standard deviation be a misleading spread measure on heavily skewed data?

Answer

It can be dominated by a handful of very large observations rather than reflecting typical variability.

#126 · Data Exploration and Visualization

Skewness statistic

Answer

A direct measure of asymmetry in a distribution; a positive value confirms a right tail suspected from a histogram.

#127 · Data Exploration and Visualization

Why is reporting the mean alongside the median a rewarded step on Exam PA?

Answer

Because a sharp divergence between them is itself evidence of skew in the underlying distribution.

#128 · Data Exploration and Visualization

Why do outliers require investigation before removal?

Answer

They can represent genuine rare events, such as a catastrophic claim, or data errors, such as a mistyped premium, and treating them the same way risks losing real information or keeping bad data.

#129 · Data Exploration and Visualization

Give the chapter's example of a genuine outlier versus a data error.

Answer

A \$50,000 medical claim in auto bodily injury may be a real severe injury claim to keep as is, while a policyholder age of 214 is almost certainly a data entry error.

#130 · Data Exploration and Visualization

What is a commonly penalized shortcut regarding outliers on Exam PA?

Answer

Simply deleting every extreme value without first investigating whether it is a genuine event or a data error.

#131 · Data Exploration and Visualization

Bivariate exploration

Answer

Looking at the relationship between two variables at once, the natural next step after understanding each variable individually.

#132 · Data Exploration and Visualization

Scatterplot

Answer

A plot of two continuous variables against each other, one on each axis, used to spot a linear or nonlinear relationship.

#133 · Data Exploration and Visualization

Give the chapter's example of a nonlinear bivariate relationship.

Answer

Claim frequency rising with age up to a point and then falling again, which suggests using a binned or nonlinear age term rather than a straight line.

#134 · Data Exploration and Visualization

Pearson correlation coefficient

Answer

A measure of the strength and direction of a linear relationship between two continuous variables, ranging from negative one to positive one.

#135 · Data Exploration and Visualization

Does a correlation near zero mean two variables have no relationship?

Answer

No, it only means there is no strong linear relationship; a scatterplot might still reveal a clear curved pattern that correlation alone would miss.

#136 · Data Exploration and Visualization

Why should a scatterplot and a correlation value be examined together?

Answer

Correlation alone can miss a nonlinear pattern that a scatterplot would reveal, so relying on either alone can give an incomplete picture.

#137 · Data Exploration and Visualization

Multicollinearity (as introduced via correlation)

Answer

A problem in a linear model when two highly correlated predictors, such as vehicle age and vehicle model year, carry largely redundant information.

#138 · Data Exploration and Visualization

Faceted or grouped plot

Answer

A chart broken into a grid of small multiples, one panel per level of a categorical variable, allowing comparison across groups at a glance.

#139 · Data Exploration and Visualization

Give the chapter's example of a faceted plot use case.

Answer

A faceted set of histograms of claim severity, one panel per policy state, to see whether some states have a fatter right tail.

#140 · Data Exploration and Visualization

Why is faceting often preferred over overlapping charts for many groups?

Answer

Overlapping histograms or scatterplots for more than two or three groups usually become unreadable, an example of too much ink obscuring the pattern.

#141 · Data Exploration and Visualization

Cross-tabulation

Answer

Also called a contingency table, it summarizes the joint distribution of two categorical variables by counting observations in each combination of categories.

#142 · Data Exploration and Visualization

Give the chapter's example of a cross-tabulation.

Answer

Coverage type (basic, standard, premium) against claim outcome (no claim, one claim, multiple claims), which can reveal self-selection into richer coverage.

#143 · Data Exploration and Visualization

Why present row or column percentages alongside raw counts in a cross-tabulation?

Answer

Because raw counts can mislead when group sizes differ substantially, such as 40 claims out of 50 basic-coverage policies versus 40 claims out of 5,000 premium-coverage policies.

#144 · Data Exploration and Visualization

What is the first step in a strong Exam PA written data exploration response?

Answer

Stating what each variable's type and role are.

#145 · Data Exploration and Visualization

What sequence of steps does a strong written Exam PA response typically follow?

Answer

State variable types and roles, evaluate time frame/sampling/granularity, produce well-chosen graphs following good design principles, and describe what the graphs show and why it matters for modeling.

#146 · Data Exploration and Visualization

What do graders consistently reward regarding exploratory findings?

Answer

Connecting an exploratory finding back to a business consequence, rather than simply stating a statistical fact and moving on.

#147 · Data Exploration and Visualization

Give the chapter's example of connecting an exploratory finding to a business consequence.

Answer

Noting that a right-skewed severity distribution suggests capping extreme claims separately in a reinsurance layer.

#148 · Data Exploration and Visualization

Is data exploration treated as a warm-up exercise on Exam PA?

Answer

No, it is graded as real analytical work in its own right, and candidates who treat it that way score better.

#149 · Data Exploration and Visualization

What does the first written task on almost every Exam PA case study involve?

Answer

Some form of data exploration and summary of the given data set.

#150 · Data Exploration and Visualization

What are graders told to reward in early exploration responses?

Answer

Candidates who describe what the data actually shows, in plain language, before jumping to modeling.

#151 · Data Exploration and Visualization

Categorize a claim severity tier of minor, moderate, severe.

Answer

It is an ordinal variable: the categories have a natural order but no consistent numeric spacing.

#152 · Data Exploration and Visualization

Categorize vehicle color.

Answer

It is a nominal variable: a category with no natural order.

#153 · Data Exploration and Visualization

Categorize annual premium.

Answer

It is a ratio variable: it has a true zero point and meaningful ratios.

#154 · Data Exploration and Visualization

Categorize driver age measured in exact years.

Answer

It is a continuous ratio variable: it can take any value within a range and has a true zero with meaningful ratios.

#155 · Data Exploration and Visualization

Categorize number of dependents.

Answer

It is a discrete ratio variable: it only takes distinct separate values and has a true zero with meaningful ratios.

#156 · Data Exploration and Visualization

Why should an actuary ask how far back a data set's time frame goes?

Answer

To check whether anything has changed in the underlying insurance product, regulation, or economic environment since then, which could make the data stale.

#157 · Data Exploration and Visualization

What question should an analyst ask about any subset of data used for modeling?

Answer

Is this the full population of relevant records, or only a slice, and if a slice, is that slice representative?

#158 · Data Exploration and Visualization

What might granularity be for a claims data set?

Answer

An individual claim transaction, an entire claim, an entire policy, or an aggregated month of experience for a book of business.

#159 · Data Exploration and Visualization

What do Exam PA graders often ask candidates to do about granularity issues?

Answer

Identify the granularity issue and propose a fix, such as aggregating claim-level data up to the policy level.

#160 · Data Exploration and Visualization

What does a clean scatterplot with a simple trend line and clear axis labels illustrate?

Answer

It communicates more than a chartjunk-laden version, such as the same plot rendered with shadowed 3D bars.

#161 · Data Exploration and Visualization

Why is a pie chart flagged as a poor choice for comparing many categories?

Answer

It becomes hard to compare slice sizes accurately as the number of categories grows, unlike a bar chart.

#162 · Data Exploration and Visualization

Why is a line chart flagged as a poor choice for unordered categories?

Answer

Connecting points that have no real ordering implies a trend or sequence that does not actually exist in the data.

#163 · Data Exploration and Visualization

What is one task Exam PA candidates are given regarding a colleague's graph?

Answer

Identifying a misleading distortion in the graph, such as a truncated axis or unlabeled log scale, and explaining why it misrepresents the data.

#164 · Data Exploration and Visualization

What is the mean's weakness as a measure of central tendency for skewed data?

Answer

It is pulled upward by a long tail of large values, so it can overstate the typical observation.

#165 · Data Exploration and Visualization

What does the whisker length in a boxplot represent?

Answer

The range of data excluding points flagged as outliers, extending from the quartiles out to the minimum and maximum non-outlier values.

#166 · Data Exploration and Visualization

What is the first quartile in a boxplot?

Answer

The value below which 25% of the observations fall.

#167 · Data Exploration and Visualization

What is the third quartile in a boxplot?

Answer

The value below which 75% of the observations fall.

#168 · Data Exploration and Visualization

Why might annual mileage and claim frequency show an upward trend in a scatterplot?

Answer

More miles driven means more exposure to the possibility of a claim occurring.

#169 · Data Exploration and Visualization

What could explain a cross-tabulation showing premium coverage associated with more claims?

Answer

Policyholders who expect to file claims may self-select into richer coverage tiers.

#170 · Data Exploration and Visualization

What insurance-relevant reason makes vehicle age and vehicle model year problematic together as predictors?

Answer

They are highly correlated and carry largely redundant information, which can create multicollinearity in a linear model.

#171 · Data Exploration and Visualization

What is one benefit of using percentages in a cross-tabulation rather than only counts?

Answer

It allows a fair comparison across groups of very different sizes.

#172 · Data Exploration and Visualization

Name a modeling implication of recognizing a right-skewed severity distribution early.

Answer

It signals a normal distribution assumption would fit poorly, pointing toward a gamma or lognormal model instead.

#173 · Data Exploration and Visualization

Which axis-related distortion involves an unlabeled transformation of scale?

Answer

Using a log scale on an axis without telling the reader, which can distort the visual impression of the underlying pattern.

#174 · Data Exploration and Visualization

What is the effect of using inconsistent bin widths on a histogram?

Answer

It can distort the visual impression of the distribution's shape, making some regions look artificially more or less dense.

#175 · Data Exploration and Visualization

How does structured data typically fit into a modeling workflow?

Answer

It fits directly into a spreadsheet or modeling data frame with no extra work, unlike unstructured data.

#176 · Data Exploration and Visualization

What is an example of a life insurance data set that is structured?

Answer

An in-force file with one row per policy and columns for issue age, face amount, and policy duration.

#177 · Data Exploration and Visualization

What kind of feature engineering step converts unstructured text into usable model inputs?

Answer

Counting keyword occurrences, flagging certain phrases, or extracting a sentiment score.

#178 · Data Exploration and Visualization

Is recognizing which data needs conversion before modeling a graded skill on Exam PA?

Answer

Yes, it is treated as a skill separate from and prior to any statistical technique.

#179 · Data Exploration and Visualization

What natural-order property distinguishes ordinal from nominal variables?

Answer

Ordinal variables have a natural order among categories, while nominal variables do not.

#180 · Data Exploration and Visualization

What true-zero property distinguishes ratio from interval variables?

Answer

Ratio variables have a true zero point that makes ratios meaningful, while interval variables do not.

#181 · Data Exploration and Visualization

Why can't you say a satisfaction rating of 'high' is twice as good as 'low'?

Answer

Because ordinal categories have no consistent numeric spacing, only a natural order.

#182 · Data Exploration and Visualization

What phrase should an actuary use to fully classify a variable for Exam PA?

Answer

State both its measurement type and its continuous or discrete nature, such as 'claim count is a discrete ratio variable.'

#183 · Data Exploration and Visualization

What role does policyholder age typically play in a lapse model?

Answer

It is a predictor variable, one of the inputs used to predict whether a policy lapses.

#184 · Data Exploration and Visualization

What role does policy lapse status play when it is the outcome being modeled?

Answer

It is the response variable, the target being predicted.

#185 · Data Exploration and Visualization

Why might a data set that only includes litigated claims mislead an actuary building a general severity model?

Answer

It excludes the ordinary, quickly settled claims, so the model overstates typical severity due to selection bias.

#186 · Data Exploration and Visualization

How does the proportional ink principle apply beyond bar charts?

Answer

Any visual encoding of a data value, such as bar height, area, or bubble size, should scale proportionally to the actual value it represents.

#187 · Data Exploration and Visualization

What is the general risk of adding heavy background gridlines to a chart?

Answer

They add visual clutter without adding information, competing with the data for the reader's attention as a form of chartjunk.

#188 · Data Exploration and Visualization

Why might redundant color legends be considered chartjunk?

Answer

When direct labels on the data itself would communicate the same information more simply, an extra legend only adds unnecessary decoration.

#189 · Data Exploration and Visualization

What does a mosaic plot help visualize?

Answer

The relationship between two categorical variables, similar to a grouped bar chart but with box areas proportional to group sizes.

#190 · Data Exploration and Visualization

Why is comparing five vehicle-class boxplots on one chart more efficient than five separate charts?

Answer

It allows a direct, at-a-glance visual comparison of spread, median, and outliers across all groups at once.

#191 · Data Exploration and Visualization

What statistical evidence would confirm a suspected right skew seen in a histogram?

Answer

A positive skewness statistic.

#192 · Data Exploration and Visualization

Why is a claim count of zero meaningful under the ratio scale?

Answer

Because zero represents a true absence of claims, which is what makes claim count a ratio rather than an interval variable.

#193 · Data Exploration and Visualization

What audience-related judgment does Exam PA reward when a case study specifies a stakeholder?

Answer

Tailoring the chart's complexity and labeling to match the stated audience in the prompt.

#194 · Data Exploration and Visualization

Why might a pandemic be relevant to evaluating a data set's time frame?

Answer

A pandemic can cause a real shift in claim frequency, so a time frame too short to capture it may miss an important pattern.

#195 · Data Exploration and Visualization

What does a scatterplot of loss ratio against policy year primarily help identify?

Answer

Whether there is a trend, linear or otherwise, in loss ratio over time.

#196 · Data Exploration and Visualization

What is the practical difference between granularity at the claim level versus the policy level?

Answer

Claim-level rows can capture multiple claims per policy, while policy-level rows aggregate all of a policyholder's claims into one summary record.

#197 · Data Exploration and Visualization

Why does an Exam PA response benefit from stating a variable's role, not just its type?

Answer

Knowing whether a variable is a response or a predictor determines how it should be used in the modeling process ahead.

#198 · Data Exploration and Visualization

What is the risk of relying only on a correlation coefficient to judge model-worthy predictors?

Answer

It can miss meaningful nonlinear relationships that a scatterplot would reveal, leading to an incomplete assessment of a variable's usefulness.

#199 · Data Exploration and Visualization

Why is a cross-tabulation especially useful before building a categorical predictor model?

Answer

It reveals the joint distribution of two categorical variables directly, showing patterns like self-selection that inform which factors to include.

#200 · Data Exploration and Visualization

What overall lesson does Section 7 give about connecting exploration to business consequences?

Answer

An exploratory finding is most valuable when tied to a concrete business action, such as capping extreme claims in a reinsurance layer, not left as a standalone statistical observation.

#201 · Data Transformations and Unsupervised Learning

What is feature engineering?

Answer

The process of creating new variables from existing data so a model can find relationships more easily, such as ratios, interaction terms, or binned versions of continuous variables.

#202 · Data Transformations and Unsupervised Learning

Why bin a continuous variable like driver age?

Answer

Binning lets a linear model assign a separate coefficient to each age group, capturing a nonlinear, U-shaped relationship with claim frequency that a single numeric coefficient cannot capture.

#203 · Data Transformations and Unsupervised Learning

What is an interaction term?

Answer

A new feature, often formed by multiplying two variables together, that lets a model capture how the effect of one predictor changes depending on the level of another predictor.

#204 · Data Transformations and Unsupervised Learning

Why should you have a story before adding an interaction term?

Answer

Adding interactions with no underlying rationale increases overfitting risk and produces a model that is harder to explain to a non-technical audience.

#205 · Data Transformations and Unsupervised Learning

Give an actuarial example of a ratio feature and why it is useful.

Answer

Loss ratio (losses divided by premium) standardizes profitability across books of business, since neither losses nor premium alone reveals whether a book is profitable.

#206 · Data Transformations and Unsupervised Learning

Why apply a log transform to claim severity?

Answer

Claim severity is typically right-skewed with a long tail of large claims; a log transform compresses that tail, making the distribution more symmetric and relationships with predictors more linear.

#207 · Data Transformations and Unsupervised Learning

What caution applies when log transforming a variable?

Answer

The log of zero or a negative number is undefined, so a small constant is sometimes added before transforming, and that choice should be explained.

#208 · Data Transformations and Unsupervised Learning

What is dummy encoding?

Answer

A method of converting a categorical variable into binary indicator variables, one for each level except a reference level, so the variable can be used in linear models and GLMs without perfect collinearity with the intercept.

#209 · Data Transformations and Unsupervised Learning

What problem can high-cardinality categorical variables cause?

Answer

A variable like ZIP code with hundreds of levels creates too many dummy variables and levels with very little data; combining rare levels or replacing the variable with a numeric summary can help.

#210 · Data Transformations and Unsupervised Learning

What is a numeric summary encoding for a high-cardinality categorical variable?

Answer

Replacing each category level with a numeric value, such as the average historical loss ratio for that level, instead of using one-hot encoding.

#211 · Data Transformations and Unsupervised Learning

What is principal components analysis (PCA)?

Answer

A technique that reduces the number of numeric variables by constructing new variables, called principal components, that are linear combinations of the originals and capture as much variance as possible.

#212 · Data Transformations and Unsupervised Learning

What defines the first principal component?

Answer

The direction in the data that captures the maximum possible variance among all possible linear combinations of the original variables.

#213 · Data Transformations and Unsupervised Learning

What constraint does the second principal component satisfy?

Answer

It captures the next most variance possible, subject to being uncorrelated (orthogonal) with the first principal component.

#214 · Data Transformations and Unsupervised Learning

Why must variables be standardized before running PCA?

Answer

PCA finds directions of maximum variance, and variance depends on measurement scale; without standardization a variable measured on a larger scale, like premium in dollars, would dominate the analysis regardless of its actual informativeness.

#215 · Data Transformations and Unsupervised Learning

What matrix does standardized PCA effectively use?

Answer

Running PCA on standardized variables is equivalent to running PCA on the correlation matrix rather than the covariance matrix.

#216 · Data Transformations and Unsupervised Learning

What is a loading in PCA?

Answer

A coefficient describing how strongly, and in what direction, each original variable contributes to a given principal component.

#217 · Data Transformations and Unsupervised Learning

How would you interpret a principal component with large positive loadings on building age and negative loadings on claims-free years?

Answer

It can be interpreted as an overall property risk axis, where older buildings with less claims-free history score high on that component.

#218 · Data Transformations and Unsupervised Learning

What is a scree plot?

Answer

A graph showing the variance explained (or cumulative variance explained) by each principal component, plotted against the component number.

#219 · Data Transformations and Unsupervised Learning

What does the elbow in a scree plot indicate?

Answer

The point where the curve of variance explained bends from steep to flat, often used as a visual rule of thumb for choosing how many principal components to keep.

#220 · Data Transformations and Unsupervised Learning

Name two common ways to choose the number of principal components to retain.

Answer

Keeping enough components to explain a target cumulative percentage of variance (such as 80 or 90 percent), or using the scree plot elbow; the Kaiser rule of keeping components with an eigenvalue above one is another option.

#221 · Data Transformations and Unsupervised Learning

Is PCA supervised or unsupervised?

Answer

Unsupervised; PCA never looks at a target variable, so it is not guaranteed to produce components that are useful predictors of the outcome you ultimately care about.

#222 · Data Transformations and Unsupervised Learning

What is a downside of using principal components as model predictors?

Answer

Components are combinations of all input variables, which can make the resulting model harder to explain to a business audience than a model built on the original, interpretable variables.

#223 · Data Transformations and Unsupervised Learning

What is a typical actuarial use case for PCA before modeling?

Answer

Reducing a large block of correlated rating variables, such as building age, roof age, and years since renovation, into a handful of components before feeding them into a GLM.

#224 · Data Transformations and Unsupervised Learning

What is clustering?

Answer

The task of grouping observations so that observations within a group are similar to each other and different from observations in other groups, with no target variable guiding the process.

#225 · Data Transformations and Unsupervised Learning

What is a common actuarial application of clustering?

Answer

Grouping policyholders into risk segments based on rating variables, without pre-specifying what the segments should look like, to support targeted underwriting or marketing.

#226 · Data Transformations and Unsupervised Learning

Describe the two repeating steps of the K-means algorithm.

Answer

Assign every observation to its nearest centroid, then recompute each centroid as the mean of all observations currently assigned to it, repeating until assignments stop changing.

#227 · Data Transformations and Unsupervised Learning

Why is K-means run multiple times from different starting centroids?

Answer

The algorithm converges to a final answer that depends on the random initial centroid placement, so multiple runs help find the result with the lowest total within-cluster variation.

#228 · Data Transformations and Unsupervised Learning

What distance metric does K-means typically use?

Answer

Euclidean distance, the straight-line distance between two points, calculated after all variables have been standardized.

#229 · Data Transformations and Unsupervised Learning

Why must variables be standardized before K-means clustering?

Answer

Because Euclidean distance is affected by scale; a variable measured on a much larger numeric scale, such as premium in dollars, would dominate the distance calculation and effectively decide the clustering alone.

#230 · Data Transformations and Unsupervised Learning

What is the elbow method used for in K-means?

Answer

Choosing the number of clusters k by plotting within-cluster sum of squares against k and looking for the point where the rate of improvement slows sharply.

#231 · Data Transformations and Unsupervised Learning

Does within-cluster sum of squares always decrease as k increases?

Answer

Yes; more clusters can always fit the data at least as well, so within-cluster sum of squares never increases as k grows, which is why the elbow shape rather than a minimum is used to choose k.

#232 · Data Transformations and Unsupervised Learning

Name two limitations of K-means clustering.

Answer

It assumes clusters are roughly spherical and similar in size, and it requires the number of clusters k to be chosen in advance rather than discovered by the algorithm.

#233 · Data Transformations and Unsupervised Learning

Why is K-means sensitive to outliers?

Answer

Because centroids are computed as means of assigned points, a single extreme outlier can pull a centroid noticeably away from the bulk of the cluster's other points.

#234 · Data Transformations and Unsupervised Learning

How should you validate K-means clusters in an actuarial context?

Answer

Check whether the resulting groups actually differ in a business-relevant way, for example by comparing average claim frequency or severity across the segments.

#235 · Data Transformations and Unsupervised Learning

What is agglomerative hierarchical clustering?

Answer

A clustering method that starts with every observation as its own cluster and repeatedly merges the two closest clusters, one merge at a time, until all observations form a single cluster.

#236 · Data Transformations and Unsupervised Learning

What is a dendrogram?

Answer

A tree diagram showing individual observations as leaves at the bottom and successive cluster merges as branches joining upward, ending in a single root.

#237 · Data Transformations and Unsupervised Learning

What does the height at which two branches join in a dendrogram represent?

Answer

How dissimilar the two merging clusters were at the point they were combined.

#238 · Data Transformations and Unsupervised Learning

What is single linkage?

Answer

A linkage method that defines the distance between two clusters as the distance between the single closest pair of points, one from each cluster; it tends to produce long, thin, chained clusters.

#239 · Data Transformations and Unsupervised Learning

What is complete linkage?

Answer

A linkage method that defines the distance between two clusters as the distance between the single farthest pair of points, one from each cluster; it tends to produce more compact, evenly sized clusters.

#240 · Data Transformations and Unsupervised Learning

What is average linkage?

Answer

A linkage method that uses the average distance across all pairs of points between two clusters, striking a middle ground between single and complete linkage.

#241 · Data Transformations and Unsupervised Learning

What is Ward's method?

Answer

A linkage method that merges the pair of clusters at each step that produces the smallest possible increase in total within-cluster variance; it tends to produce similarly sized clusters and is a common default.

#242 · Data Transformations and Unsupervised Learning

How do you obtain a specific set of clusters from a dendrogram?

Answer

By cutting the tree: drawing a horizontal line at a chosen height and treating every branch the line crosses as a separate cluster.

#243 · Data Transformations and Unsupervised Learning

How does the cut height affect the resulting clusters?

Answer

Cutting higher on the dendrogram produces fewer, larger clusters, while cutting lower produces more, smaller clusters.

#244 · Data Transformations and Unsupervised Learning

What visual cue can justify a particular dendrogram cut height?

Answer

A large vertical gap between successive merges, implying the next merge would combine clusters that are genuinely quite different.

#245 · Data Transformations and Unsupervised Learning

What is a key advantage of hierarchical clustering over K-means?

Answer

Hierarchical clustering does not require specifying the number of clusters in advance and produces a dendrogram showing the full nested merge structure at every level of granularity.

#246 · Data Transformations and Unsupervised Learning

What is a key advantage of K-means over hierarchical clustering?

Answer

K-means is generally faster and more scalable on large datasets, since it does not need to compute and store distances between every pair of observations.

#247 · Data Transformations and Unsupervised Learning

Which linkage methods are more robust to a single extreme outlier?

Answer

Complete linkage and Ward's method tend to be more robust to a single extreme point than single linkage.

#248 · Data Transformations and Unsupervised Learning

Why is it essential to interpret clusters after either K-means or hierarchical clustering?

Answer

Neither method uses a target variable, so clusters must be checked against business measures like claim frequency, severity, or retention rate to confirm they carry real business meaning.

#249 · Data Transformations and Unsupervised Learning

Why might binning also help with a high-cardinality variable like vehicle make?

Answer

Grouping the rarest makes into an 'other' category prevents a model from trying to estimate a coefficient from only a handful of observations.

#250 · Data Transformations and Unsupervised Learning

What is one reason a log-transformed response might fit insurance severity data better than the raw dollar amount?

Answer

Many models assume roughly symmetric, homoscedastic errors, and log transforming a right-skewed severity variable brings its distribution closer to that assumption.

#251 · Data Transformations and Unsupervised Learning

What kind of variable is loss ratio, and why is it useful as a feature?

Answer

It is a ratio feature (losses divided by premium) that captures relative profitability, which neither raw losses nor raw premium can show on its own.

#252 · Data Transformations and Unsupervised Learning

How can a ratio feature reduce collinearity?

Answer

Replacing two correlated raw variables with a single ratio, such as claims per year of exposure, can produce a more interpretable measure with less redundancy than keeping both raw variables.

#253 · Data Transformations and Unsupervised Learning

What general reasoning structure does Exam PA reward for feature engineering proposals?

Answer

State the problem observed in the raw data, propose a specific transformation, and explain why that transformation addresses the problem for the business question at hand.

#254 · Data Transformations and Unsupervised Learning

What happens to the number of principal components as you keep computing them?

Answer

You can compute as many principal components as there are original variables, though in practice only the first few, which capture most of the variance, are retained.

#255 · Data Transformations and Unsupervised Learning

What is the Kaiser rule for choosing principal components?

Answer

A rule of thumb that keeps only components with an eigenvalue greater than one when PCA is run on a standardized correlation matrix.

#256 · Data Transformations and Unsupervised Learning

Why is PCA often described as exploratory rather than a final modeling answer?

Answer

Because it does not use the target variable and produces components that can be hard to interpret directly, it is often used as a preprocessing step before a supervised model rather than the final analysis.

#257 · Data Transformations and Unsupervised Learning

What is the effect of choosing k too small in K-means?

Answer

Distinct groups in the data may be merged together, hiding meaningful business differences between segments that a larger k would have separated.

#258 · Data Transformations and Unsupervised Learning

What is the effect of choosing k too large in K-means?

Answer

The clusters may become overly fine and harder to interpret or act on operationally, even though within-cluster sum of squares keeps decreasing.

#259 · Data Transformations and Unsupervised Learning

Why might a business context override a purely statistical choice of k ?

Answer

A company may only be able to operationally support a fixed number of rating tiers, making a smaller or specific k more practical even if the elbow method suggests otherwise.

#260 · Data Transformations and Unsupervised Learning

What shape assumption does K-means make about clusters?

Answer

It assumes clusters are roughly spherical and similar in size, so it performs poorly on elongated, irregularly shaped, or very differently sized true groupings.

#261 • Data Transformations and Unsupervised Learning

What computational drawback does hierarchical clustering have compared to K-means as data grows?

Answer

It is generally more computationally expensive, since a full hierarchical merge requires repeated distance calculations across all pairs of clusters.

#262 • Data Transformations and Unsupervised Learning

When might hierarchical clustering be preferred over K-means?

Answer

For smaller datasets or exploratory work where you want to see multiple candidate groupings from one dendrogram before committing to a final number of clusters.

#263 · Data Transformations and Unsupervised Learning

What does it mean that the first and second principal components are orthogonal?

Answer

They are uncorrelated with each other, meaning each successive component captures variance not already explained by the earlier components.

#264 · Data Transformations and Unsupervised Learning

Give an example of an interaction term in a homeowners insurance model.

Answer

Building age multiplied by an indicator for outdated wiring, capturing that the effect of building age on claim severity is stronger for homes without updated wiring.

#265 · Data Transformations and Unsupervised Learning

What is a reference level in dummy encoding?

Answer

The one category level left out of the set of dummy variables; coefficients on the other dummy variables are interpreted relative to this baseline level.

#266 · Data Transformations and Unsupervised Learning

Why might tree-based models handle categorical variables differently than linear models?

Answer

Tree-based models can often split directly on category levels without one-hot encoding, though many software implementations still expect encoded input.

#267 · Data Transformations and Unsupervised Learning

What is the practical effect of standardizing before both PCA and K-means?

Answer

It ensures no single variable dominates the analysis purely due to its measurement scale, so variance (in PCA) or distance (in K-means) reflects genuine patterns rather than units.

#268 · Data Transformations and Unsupervised Learning

What does cumulative variance explained mean in PCA?

Answer

The running total proportion of the dataset's total variance captured by the first several principal components combined, used to decide how many components to retain.

#269 · Data Transformations and Unsupervised Learning

What is a policyholder risk-segmentation use case that combines PCA and clustering?

Answer

Using PCA to reduce a set of correlated rating variables to a few components, then clustering policyholders on those components to define risk segments with less redundant, standardized input.

#270 · Data Transformations and Unsupervised Learning

Why is it risky to add many interaction terms without justification?

Answer

Each added interaction increases model complexity and overfitting risk while reducing interpretability, so interactions should be limited to ones with a clear underlying rationale.

#271 · Data Transformations and Unsupervised Learning

What is exposure, and why divide claim count by it?

Answer

Exposure measures how long a policyholder has been at risk, such as years on the books; dividing claim count by exposure standardizes claim frequency across policyholders with different exposure lengths.

#272 · Data Transformations and Unsupervised Learning

How does a dendrogram let you avoid rerunning an algorithm for every candidate cluster count?

Answer

A single dendrogram encodes every possible number of clusters at once, since cutting it at different heights yields different numbers of clusters without recomputing the merge structure.

#273 · Data Transformations and Unsupervised Learning

What is one reason a candidate should always interpret PCA loadings in business language on Exam PA?

Answer

A component is just a set of numbers until its loadings are translated into a meaningful business concept, such as an overall property risk axis, which graders specifically look for.

#274 · Data Transformations and Unsupervised Learning

Why is Ward's method often a sensible default linkage choice?

Answer

It tends to produce clusters that are similar in size and works well when the underlying true groups are expected to be roughly compact, unlike single linkage's tendency to chain.

#275 · Data Transformations and Unsupervised Learning

What single validation step applies to both PCA-derived features and cluster assignments before using them in an actuarial project?

Answer

Confirming the resulting features or groups actually relate to a business outcome of interest, such as loss ratio or claim frequency, rather than trusting statistical output alone.

#276 · Generalized Linear Models

What are the three components of a GLM?

Answer

A random component specifying the response distribution, a systematic component (the linear predictor), and a link function connecting the mean of the response to the linear predictor.

#277 · Generalized Linear Models

Why is ordinary linear regression a poor fit for claim counts?

Answer

Claim counts are non-negative integers with variance that changes with the mean, while ordinary regression assumes a continuous response with constant variance and can predict negative values.

#278 · Generalized Linear Models

Which distribution is standard for modeling claim frequency?

Answer

The Poisson distribution, since it models non-negative counts and assumes the variance equals the mean.

#279 · Generalized Linear Models

What does it mean for a Poisson model to be overdispersed?

Answer

The observed variance in the data exceeds what the mean predicts, so the simple Poisson assumption of variance equal to mean no longer holds.

#280 · Generalized Linear Models

Name two fixes for an overdispersed Poisson frequency model.

Answer

Fit a quasi-Poisson model that scales the variance by an estimated dispersion parameter, or fit a negative binomial model that adds a separate parameter for the extra variance.

#281 · Generalized Linear Models

Which distribution is standard for modeling claim severity?

Answer

The gamma distribution, because it is continuous, strictly positive, right-skewed, and has variance that grows with the square of the mean.

#282 · Generalized Linear Models

What property of the gamma distribution matches insurance severity data?

Answer

A constant coefficient of variation, meaning the standard deviation grows proportionally with the mean, matching the pattern that larger expected claims have more absolute variability.

#283 · Generalized Linear Models

When would an actuary choose the inverse Gaussian distribution over gamma for severity?

Answer

When the severity distribution has an even heavier tail than gamma implies, since inverse Gaussian variance grows with the cube of the mean rather than the square.

#284 · Generalized Linear Models

What business quantity is pure premium?

Answer

Loss cost per unit of exposure, equal to expected claim frequency multiplied by expected claim severity.

#285 · Generalized Linear Models

Why is pure premium data difficult to model with a single standard distribution?

Answer

Many policies have exactly zero losses, creating a point mass at zero, while policies with losses follow a continuous positive right-skewed distribution.

#286 · Generalized Linear Models

What distribution is used to model pure premium directly?

Answer

The Tweedie distribution, which handles the point mass at zero combined with a continuous positive tail in one model.

#287 · Generalized Linear Models

What is the Tweedie power parameter p , and what do its endpoints represent?

Answer

A parameter usually between 1 and 2 controlling the distribution shape; p equal to 1 reduces Tweedie to Poisson, and p equal to 2 reduces it to gamma.

#288 · Generalized Linear Models

What underlying process does a Tweedie distribution with p strictly between 1 and 2 represent?

Answer

A compound Poisson-gamma process, a Poisson-distributed number of claims each with a gamma-distributed severity, summed together.

#289 · Generalized Linear Models

What does a Tweedie p value closer to 1 imply about the data?

Answer

More Poisson-like behavior, meaning more probability mass concentrated at exactly zero.

#290 · Generalized Linear Models

Why has Tweedie regression become popular for pure premium modeling?

Answer

It lets an actuary build a single pure premium model directly from the data rather than fitting separate frequency and severity models and multiplying the results.

#291 · Generalized Linear Models

Which distribution and link are standard for modeling a binary outcome like policy lapse?

Answer

The binomial distribution with a logit link, giving logistic regression.

#292 · Generalized Linear Models

How does binomial variance behave as the probability changes?

Answer

Variance equals the mean times one minus the mean, so it is largest near a probability of 0.5 and shrinks toward 0 or 1.

#293 · Generalized Linear Models

What is the formula for the log link function?

Answer

η of μ equals the natural log of μ , connecting the mean of the response to the linear predictor.

#294 · Generalized Linear Models

Why does a log link guarantee positive predictions?

Answer

Predictions are recovered by exponentiating the linear predictor, and the exponential function is always positive regardless of the input value.

#295 · Generalized Linear Models

Why is a log link called multiplicative rather than additive?

Answer

Because the prediction is $\exp$ of a sum of terms, which equals a product of $\exp$ of each individual term, so each variable contributes a multiplicative factor.

#296 · Generalized Linear Models

What is the formula for the logit link function?

Answer

η of μ equals the natural log of μ divided by one minus μ , mapping a probability between 0 and 1 onto the entire real line.

#297 · Generalized Linear Models

What does exponentiating a logistic regression coefficient produce?

Answer

An odds ratio, the multiplicative change in the odds of the outcome associated with a one-unit change in that predictor.

#298 · Generalized Linear Models

What is the identity link, and when is it appropriate?

Answer

$g(\mu) = \mu$, the ordinary regression link; it is rarely appropriate for insurance loss data since it allows negative predictions.

#299 · Generalized Linear Models

Why is a log link preferred over an identity link for severity models?

Answer

Severity must remain strictly positive, and only the log link guarantees positive fitted values after back-transformation.

#300 · Generalized Linear Models

What is the purpose of an offset in a GLM?

Answer

To scale the response so the model predicts a rate per unit of exposure rather than a raw total, without estimating a coefficient for it.

#301 · Generalized Linear Models

In a log-link Poisson frequency model, what offset term is added and what is its coefficient?

Answer

The natural log of exposure is added to the linear predictor with its coefficient fixed at exactly 1, not estimated from the data.

#302 · Generalized Linear Models

If a policy is in force for 0.5 of a year instead of a full year, how does an offset model treat expected claims?

Answer

It expects about half as many claims as an identical policy in force a full year, all else equal, since claim count scales with exposure.

#303 · Generalized Linear Models

What happens to the model if exposure is left out entirely, with no offset or predictor?

Answer

Exposure differences between policies contaminate the estimated effects of every other explanatory variable in the model.

#304 · Generalized Linear Models

What is the purpose of a weight in a GLM, as distinct from an offset?

Answer

A weight does not change what the response represents; it changes how much influence each observation has on parameter estimation by scaling the assumed variance.

#305 · Generalized Linear Models

When should an actuary use a weight rather than an offset?

Answer

When the response variable is already an average or a rate, such as average claim severity, but different records are based on different amounts of underlying data.

#306 · Generalized Linear Models

In a severity model where each record is an average claim size, what is typically used as the weight?

Answer

The number of claims backing up that average, so classes with more claims supporting their average get more influence in fitting.

#307 · Generalized Linear Models

What is the key discriminating question for choosing an offset versus a weight?

Answer

Does the response variable represent a total, which calls for an offset, or an average or rate, which calls for a weight?

#308 · Generalized Linear Models

Under a log link, what does a fitted coefficient represent before exponentiation?

Answer

The additive change in the log of the mean response associated with a one-unit change in that predictor, holding other predictors fixed.

#309 · Generalized Linear Models

What does exponentiating a log-link coefficient produce?

Answer

A multiplicative relativity applied to the baseline predicted value.

#310 · Generalized Linear Models

If a Territory coefficient for Urban is 0.3247 in a log-link Poisson model, what is the Urban relativity?

Answer

exp of 0.3247, which equals approximately 1.3836, meaning Urban policies have about 1.3836 times the frequency of Rural policies.

#311 · Generalized Linear Models

If the baseline Rural frequency is 0.0800 claims per exposure and the Urban relativity is 1.3836, what is the fitted Urban frequency?

Answer

0.0800 times 1.3836, which equals approximately 0.1107 claims per exposure.

#312 · Generalized Linear Models

What relativity does a coefficient of exactly zero always exponentiate to?

Answer

A relativity of exactly 1, meaning no effect relative to the baseline level.

#313 · Generalized Linear Models

What relativity is always implied by the reference level of a categorical variable?

Answer

A relativity of 1.000, even though the reference level never appears explicitly in the coefficient table.

#314 · Generalized Linear Models

What sign of coefficient always produces a relativity above 1 after exponentiation?

Answer

A positive coefficient, since exp of any positive number is greater than 1.

#315 · Generalized Linear Models

What sign of coefficient always produces a relativity below 1 after exponentiation?

Answer

A negative coefficient, since exp of any negative number is between 0 and 1.

#316 · Generalized Linear Models

How should combined main effects and an interaction term be handled before exponentiating?

Answer

Add the relevant coefficients together at the linear predictor scale first, then exponentiate the sum, rather than exponentiating each piece separately and multiplying.

#317 · Generalized Linear Models

If Urban is 0.3247, Young Driver is 0.1823, and their interaction is negative 0.0950, what is the combined Urban Young Driver relativity?

Answer

exp of (0.3247 plus 0.1823 minus 0.0950), which is exp of 0.4120, approximately 1.5099.

#318 · Generalized Linear Models

Why does a negative interaction coefficient reduce the combined relativity below the product of the two main-effect relativities?

Answer

It captures that having both risk factors together does not compound risk as strongly as the two main effects would separately suggest.

#319 · Generalized Linear Models

When does multiplying two separately exponentiated main-effect relativities give the same answer as exponentiating the summed coefficients?

Answer

Only when the model contains no interaction term between those two variables.

#320 · Generalized Linear Models

In a logistic regression with a coefficient of 0.5108 on a prior-claim indicator, what is the odds ratio?

Answer

exp of 0.5108, approximately 1.6667, meaning the odds of lapsing are about 1.6667 times higher with a recent prior claim.

#321 · Generalized Linear Models

What is the key distinction between an odds ratio and a probability change?

Answer

An odds ratio describes a multiplicative change in the odds of an outcome, not a direct multiplicative change in the probability itself.

#322 · Generalized Linear Models

What problem does regularized regression address?

Answer

Overfitting and unstable coefficients that arise from maximum likelihood estimation when a model has many, possibly correlated, predictors.

#323 · Generalized Linear Models

What penalty does ridge regression add to the objective function?

Answer

A penalty proportional to the sum of the squared coefficients.

#324 · Generalized Linear Models

Does ridge regression perform variable selection?

Answer

No, ridge shrinks all coefficients toward zero but essentially never sets any coefficient to exactly zero.

#325 · Generalized Linear Models

What penalty does lasso regression add to the objective function?

Answer

A penalty proportional to the sum of the absolute values of the coefficients.

#326 · Generalized Linear Models

What makes lasso useful for variable selection?

Answer

Its penalty shape can shrink some coefficients all the way to exactly zero, effectively removing those variables from the model.

#327 · Generalized Linear Models

What is elastic net regression?

Answer

A method combining both the ridge and lasso penalties, using a mixing parameter that controls the balance between the two.

#328 · Generalized Linear Models

In elastic net, what does an alpha of 1 correspond to, and what does an alpha of 0 correspond to?

Answer

Alpha of 1 is pure lasso, and alpha of 0 is pure ridge.

#329 · Generalized Linear Models

Why might elastic net be preferred over pure lasso when predictors are correlated?

Answer

Pure lasso tends to arbitrarily pick one variable from a correlated group and zero out the rest, while elastic net tends to keep or drop correlated groups together.

#330 · Generalized Linear Models

What hyperparameter controls the strength of a regularization penalty?

Answer

Lambda, which is not estimated by maximum likelihood alongside the coefficients and must be chosen separately.

#331 · Generalized Linear Models

What method is standard for choosing the regularization hyperparameter λ ?

Answer

K-fold cross-validation, selecting the λ that minimizes average validation error across the folds.

#332 · Generalized Linear Models

What model results when λ equals exactly zero in a regularized regression?

Answer

The unregularized model, since the penalty term contributes nothing.

#333 · Generalized Linear Models

What happens to a regularized model as λ grows very large?

Answer

Coefficients are pushed toward zero and the model becomes simpler, eventually approaching an intercept-only model.

#334 · Generalized Linear Models

Why might an actuary choose a λ slightly larger than the one that minimizes cross-validation error?

Answer

A larger λ within one standard error of the minimum often yields a simpler, more interpretable model at nearly the same predictive accuracy.

#335 · Generalized Linear Models

How does k-fold cross-validation work?

Answer

The data is split into k roughly equal folds; the model is trained k times, each time leaving out one fold for validation, and validation error is averaged across all k runs.

#336 · Generalized Linear Models

Why is a train and test split alone sometimes insufficient for model validation?

Answer

It sacrifices a large portion of the data as a held-out test set, which can be wasteful when the overall data set is not large.

#337 · Generalized Linear Models

What advantage does k-fold cross-validation have over a single train and test split?

Answer

It uses every observation for both training and validation at some point, without ever validating a model run on data used to fit that same run.

#338 · Generalized Linear Models

What is deviance in the context of a GLM?

Answer

The standard goodness-of-fit measure for a GLM, generalizing residual sum of squares to any exponential family distribution, where lower deviance indicates better fit.

#339 · Generalized Linear Models

Why can deviance alone not be used to compare models with different numbers of predictors?

Answer

Deviance almost always decreases when a variable is added, even a useless one, so it rewards complexity regardless of true improvement.

#340 · Generalized Linear Models

What is AIC, and what does it add to a model's fit measure?

Answer

The Akaike Information Criterion, which adds a penalty for the number of parameters used so added complexity must earn its place.

#341 · Generalized Linear Models

What is BIC, and how does its penalty differ from AIC's?

Answer

The Bayesian Information Criterion; its penalty per parameter grows with the natural log of the sample size, penalizing extra parameters more heavily than AIC in most realistic sample sizes.

#342 · Generalized Linear Models

When AIC and BIC disagree on model selection, which tends to favor a smaller model?

Answer

BIC, because of its heavier penalty per additional parameter.

#343 · Generalized Linear Models

How should a candidate select among several candidate GLMs using AIC or BIC?

Answer

Choose the model with the lowest AIC or BIC value, since lower values indicate a preferred balance of fit and complexity.

#344 · Generalized Linear Models

What is a lift chart?

Answer

A graphical validation tool where test-set policies are sorted by predicted value, grouped into buckets such as deciles, and average predicted versus actual values are plotted per bucket.

#345 · Generalized Linear Models

What does a well-performing lift chart look like?

Answer

Predicted and actual values track closely together and rise steadily and monotonically from the lowest bucket to the highest.

#346 · Generalized Linear Models

What is a double lift chart used for?

Answer

Comparing two competing models directly against each other, rather than each against actual data separately.

#347 · Generalized Linear Models

How are policies grouped in a double lift chart?

Answer

By the ratio of one model's prediction to the other model's prediction, then grouped into buckets, with actual losses plotted alongside both models' predictions.

#348 · Generalized Linear Models

In a double lift chart, how is the better model identified?

Answer

The model whose predicted curve tracks the actual curve more closely across the buckets, especially at the extremes, is the better-differentiating model.

#349 · Generalized Linear Models

What does the Gini index measure?

Answer

A model's ability to rank-order risks correctly, summarized as a single number between 0 and 1, with higher values indicating better discrimination.

#350 · Generalized Linear Models

How is the Gini index typically constructed?

Answer

By sorting policies by predicted risk and plotting cumulative share of actual losses captured against cumulative share of exposure.

#351 · Generalized Linear Models

What is the exponential family, and why does it matter for GLMs?

Answer

A class of probability distributions including Poisson, gamma, binomial, and Tweedie, whose shared mathematical structure allows a single unified GLM fitting framework.

#352 · Generalized Linear Models

What is the linear predictor in a GLM, commonly denoted η ?

Answer

The intercept plus the sum of coefficients times explanatory variables, before any link transformation is applied.

#353 · Generalized Linear Models

How is a fitted mean μ recovered from the linear predictor η under a log link?

Answer

μ equals $\exp(\eta)$, since the log link states that $\ln(\mu)$ equals η .

#354 · Generalized Linear Models

In an offset model where $\ln(\mu)$ equals $\ln(\text{exposure})$ plus η , what does $\exp(\eta)$ represent?

Answer

The expected claim frequency per unit of exposure, which is the quantity actuaries actually want to price.

#355 · Generalized Linear Models

Why does a gamma GLM with claim-count weights give the same estimates as fitting to individual claim-level data?

Answer

Weighting by claim count effectively reconstructs the information content of the disaggregated data, a property relied on when only aggregated exhibits are available.

#356 · Generalized Linear Models

For an auto insurer studying claims per policy-year, which distribution and link are typical?

Answer

Poisson distribution with a log link, since the response is a non-negative count and predictions must stay positive.

#357 · Generalized Linear Models

For a workers compensation insurer modeling average indemnity claim cost, which distribution and link are typical?

Answer

Gamma distribution with a log link, since severity is continuous, strictly positive, and right-skewed.

#358 · Generalized Linear Models

What diagnostic suggests a Poisson frequency model is overdispersed?

Answer

A variance-to-mean ratio well above 1 in the observed data compared to what a plain Poisson model assumes.

#359 · Generalized Linear Models

What is a relativity, in the context of an actuarial rating plan?

Answer

A single multiplicative factor applied to a base rate to reflect the effect of one rating variable's level.

#360 · Generalized Linear Models

Why is a log-link GLM particularly well suited to building a rating algorithm?

Answer

Its multiplicative structure lets each rating variable's effect be expressed as a relativity, matching how base rates and factors are communicated to regulators and underwriters.

#361 · Generalized Linear Models

What is the reference level of a categorical predictor in a GLM?

Answer

The baseline level against which all other levels of that variable are compared, absorbed into the model's intercept.

#362 · Generalized Linear Models

Why does the exam favor Tweedie regression over separate frequency and severity models for pure premium tasks?

Answer

Tweedie builds one model directly from the pure premium response, avoiding the extra step of fitting two models and multiplying their predictions together.

#363 · Generalized Linear Models

What kind of variable is claim frequency, in terms of data type?

Answer

A count variable, taking non-negative integer values such as 0, 1, 2, and so on.

#364 · Generalized Linear Models

What kind of variable is claim severity, in terms of data type?

Answer

A continuous, strictly positive variable, typically right-skewed with a long tail of large losses.

#365 · Generalized Linear Models

What kind of variable is a fraud flag, in terms of data type?

Answer

A binary variable, taking only the values 0 or 1.

#366 · Generalized Linear Models

How does negative binomial regression differ from Poisson regression in its variance assumption?

Answer

Negative binomial adds a separate dispersion parameter so variance can exceed the mean, while Poisson forces variance to equal the mean exactly.

#367 · Generalized Linear Models

What does a quasi-Poisson model do differently from a standard Poisson model?

Answer

It keeps the same mean structure as Poisson but scales the variance estimate by an estimated dispersion parameter to account for overdispersion.

#368 · Generalized Linear Models

What is the practical consequence of choosing gamma over inverse Gaussian for a severity model with a moderately heavy tail?

Answer

Gamma assumes variance grows with the square of the mean, which is usually adequate unless the tail is unusually heavy, where inverse Gaussian's cubic variance growth fits better.

#369 · Generalized Linear Models

Why can a Tweedie model handle policies with exactly zero losses without a separate zero-inflation component?

Answer

The Tweedie distribution places genuine probability mass exactly at zero when p is between 1 and 2, matching policies with no claims at all.

#370 · Generalized Linear Models

In business terms, what does an offset let a candidate say about a fitted frequency model?

Answer

That the model predicts claims per unit of exposure directly, rather than raw claim totals that depend on how long each policy was in force.

#371 · Generalized Linear Models

Why should exposure never be entered as an ordinary predictor with an estimated coefficient in a frequency model?

Answer

Because the relationship between claims and exposure is definitionally proportional, and forcing the model to estimate that coefficient wastes a degree of freedom on a relationship already known to be 1 to 1.

#372 · Generalized Linear Models

What happens to model interpretation when weights, rather than an offset, are used incorrectly for a raw count response?

Answer

The model would treat the response as already representing a rate when it is really a total, misestimating the effect of every explanatory variable.

#373 · Generalized Linear Models

How do you convert a fitted log-link coefficient into a percentage change interpretation?

Answer

Exponentiate the coefficient to get the relativity, then subtract 1 and multiply by 100 to express it as a percentage increase or decrease from baseline.

#374 · Generalized Linear Models

If a coefficient of 0.1 gives a relativity of $\exp(0.1)$, approximately 1.105, what percentage increase does this represent?

Answer

Approximately a 10.5 percent increase over the baseline level.

#375 · Generalized Linear Models

Why do actuaries prefer expressing GLM output as relativities rather than raw coefficients when presenting to underwriters?

Answer

Relativities are directly interpretable as multiplicative rating factors, matching how underwriters and rating manuals already express risk adjustments.

#376 · Generalized Linear Models

What is the effect of adding an interaction term on the main effect coefficients already in the model?

Answer

The main effect coefficients now represent the effect of that variable only when the other interacting variable is at its reference level, not an average effect across all levels.

#377 · Generalized Linear Models

Why might a candidate include an interaction term between Territory and Vehicle Type in an auto frequency model?

Answer

If the effect of vehicle type on frequency genuinely differs by territory, a model without the interaction would misstate the combined risk for some Territory and Vehicle Type combinations.

#378 · Generalized Linear Models

What is a common mistake candidates make with interaction terms during the written exam?

Answer

Exponentiating each main effect separately and multiplying, instead of summing the coefficients including the interaction term first and then exponentiating.

#379 · Generalized Linear Models

What does regularization trade off to improve predictive performance?

Answer

A small amount of added bias in exchange for a meaningful reduction in variance, usually improving performance on new, unseen data.

#380 · Generalized Linear Models

Why can ordinary maximum likelihood estimation become unstable with many correlated predictors?

Answer

Correlated predictors make it hard to separate their individual effects, inflating standard errors and producing coefficients that vary wildly with small changes in the data.

#381 · Generalized Linear Models

What is the practical effect of choosing a very small λ in a regularized model?

Answer

The model behaves close to the unregularized fit, retaining nearly all variables with minimal shrinkage.

#382 · Generalized Linear Models

What is the practical effect of choosing a very large λ in a regularized model?

Answer

Nearly all coefficients are shrunk close to zero, producing an overly simple model that may underfit the data.

#383 · Generalized Linear Models

Why is choosing lambda by minimizing training error alone a mistake?

Answer

A lambda of zero always minimizes training error, since it recovers the unregularized model, which defeats the purpose of regularization; lambda must be chosen using held-out validation performance.

#384 · Generalized Linear Models

What does the one-standard-error rule accomplish when selecting lambda from a cross-validation curve?

Answer

It selects a somewhat larger, simpler lambda whose validation error is still within one standard error of the true minimum, favoring interpretability without meaningfully sacrificing accuracy.

#385 · Generalized Linear Models

Why is a held-out test set essential before reporting a final model's performance?

Answer

Metrics computed on training data overstate performance because the model has already seen and adapted to that data; a held-out test set was never used in fitting.

#386 · Generalized Linear Models

What is the relationship between deviance and the residual sum of squares used in ordinary regression?

Answer

Deviance generalizes residual sum of squares to any exponential family distribution, reducing to it exactly in the normal distribution case.

#387 · Generalized Linear Models

If Model A has lower deviance but more parameters than Model B, is Model A automatically better?

Answer

Not necessarily; lower deviance alone does not account for the added complexity, so AIC or BIC should be used to compare them fairly.

#388 · Generalized Linear Models

What information does a lift chart provide that AIC or BIC does not?

Answer

A direct, visual, business-relevant check of how well the model separates low-risk from high-risk policies in actual held-out experience.

#389 · Generalized Linear Models

In a lift chart, what does it mean if the actual line is flat across buckets while the predicted line rises steeply?

Answer

The model believes it is separating risk levels well, but the actual experience shows the model's ranking does not correspond to real differences in outcomes, a sign of poor calibration.

#390 · Generalized Linear Models

Why is a double lift chart specifically useful when replacing a model already in production?

Answer

It directly shows whether the proposed new model differentiates risk better than the existing model, which is the exact justification needed to support a model change.

#391 · Generalized Linear Models

What does a Gini index of 0 indicate about a model?

Answer

No discriminatory power at all, equivalent to random ranking of policies by risk.

#392 · Generalized Linear Models

What does a Gini index closer to 1 indicate about a model?

Answer

Strong discriminatory power, meaning the model ranks high-risk and low-risk policies very accurately.

#393 · Generalized Linear Models

Why might an actuary use the Gini index alongside a lift chart rather than instead of one?

Answer

The Gini index gives a single summary number for quick comparison across many candidate models, while the lift chart shows the full shape of performance across risk buckets.

#394 · Generalized Linear Models

How should a candidate describe a GLM's response distribution choice in a written exam answer for a non-technical audience?

Answer

In plain terms tied to the business problem, for example explaining that claim counts are modeled with Poisson because counts cannot be negative and vary in a specific, predictable way.

#395 · Generalized Linear Models

Why does the real Exam PA rarely ask candidates to compute GLM output entirely from scratch?

Answer

Because the exam provides R or Excel output such as coefficient tables and lift charts already generated, and instead tests interpretation and business judgment.

#396 · Generalized Linear Models

What is the most important translation skill tested by GLM tasks on the real exam?

Answer

Converting a fitted statistical model's output into a concrete, defensible business recommendation understandable by a non-technical audience.

#397 · Generalized Linear Models

Give an example of a binary response variable commonly modeled with logistic regression in insurance.

Answer

Whether a policyholder lapses their policy within a given renewal period.

#398 · Generalized Linear Models

Give an example of a Tweedie modeling application in insurance.

Answer

Modeling homeowners pure premium directly, where many policies have zero losses and others have a continuous positive loss amount.

#399 · Generalized Linear Models

What is the practical reason a candidate should always check whether exposure varies across policies before fitting a frequency model?

Answer

Unequal exposure periods across policies must be accounted for, typically via an offset, or the model's estimated effects for other variables will be biased.

#400 · Generalized Linear Models

How does adding a weight change the effective variance assumed for an observation in a GLM?

Answer

A higher weight reduces the assumed variance for that observation, giving it more influence on the fitted coefficients relative to observations with lower weights.

#401 · Generalized Linear Models

What is a candidate risk of relying only on AIC without also checking a lift chart?

Answer

AIC measures statistical fit adjusted for complexity but does not directly show whether the model actually separates risk well in a business-relevant, visual sense.

#402 · Generalized Linear Models

Why is deviance alone insufficient even when comparing models with the same number of parameters?

Answer

It is not insufficient in that case; deviance alone is a valid comparison only when the number of parameters is equal, since no complexity adjustment is then needed.

#403 · Generalized Linear Models

What role does cross-validation play beyond choosing regularization hyperparameters?

Answer

It also provides an honest estimate of how any candidate model, regularized or not, will perform on new data before final model selection.

#404 · Generalized Linear Models

Why would an actuary choose ridge regression over lasso when most predictors are believed to be genuinely relevant?

Answer

Ridge stabilizes and shrinks all coefficients without eliminating any variable, which is preferable when there is no strong reason to believe any predictor is truly useless.

#405 • Generalized Linear Models

Why would an actuary choose lasso over ridge when many candidate predictors are suspected to be irrelevant?

Answer

Lasso can shrink irrelevant predictors' coefficients all the way to zero, automatically producing a simpler, more interpretable model.

#406 • Generalized Linear Models

What is the practical business benefit of a sparser model produced by lasso or elastic net?

Answer

Fewer variables make the rating plan easier to explain to regulators, underwriters, and other stakeholders, and easier to maintain over time.

#407 · Generalized Linear Models

In the Territory example, what is the relativity if the fitted coefficient for a level were exactly negative 0.2231?

Answer

exp of negative 0.2231, approximately 0.8000, meaning that level has 80 percent of the baseline frequency.

#408 · Generalized Linear Models

Why is it important to hold all other variables fixed when interpreting a single coefficient in a GLM?

Answer

GLM coefficients describe the isolated effect of one variable controlling for the others in the model, not the raw unconditional relationship between that variable and the response.

#409 · Generalized Linear Models

What kind of exam question typically follows a coefficient table listing several categorical variable levels?

Answer

A question asking the candidate to calculate the combined fitted rate or relativity for a specific policy profile by summing the relevant coefficients and exponentiating.

#410 · Generalized Linear Models

How should the reference level's relativity be treated when calculating a combined relativity for a policy at all baseline levels?

Answer

It contributes a relativity of exactly 1 for every variable at its reference level, so the baseline policy's total relativity is 1, equal to the model's baseline rate.

#411 · Generalized Linear Models

What is the danger of interpreting a Tweedie model's fitted value as if it always represents a nonzero claim?

Answer

The fitted value is an expected pure premium averaged across all outcomes including the possibility of zero loss, not a prediction that a claim will definitely occur.

#412 · Generalized Linear Models

Why might a workers compensation actuary prefer a gamma severity model with claim-count weights over an unweighted model?

Answer

Unweighted fitting would let a class with very few claims and a noisy average severity distort the fitted relationship as much as a well-supported class with many claims.

#413 · Generalized Linear Models

What distinguishes a systematic component from a random component in a GLM?

Answer

The systematic component is the linear predictor built from explanatory variables and coefficients, while the random component specifies the probability distribution assumed for the response.

#414 · Generalized Linear Models

How does a candidate justify choosing logistic regression over linear regression for a claim fraud model?

Answer

The outcome is binary, so logistic regression's logit link keeps predicted probabilities between 0 and 1, unlike linear regression which can predict values outside that range.

#415 · Generalized Linear Models

What happens to a fitted probability from a logistic regression as the linear predictor approaches very large positive values?

Answer

The fitted probability approaches 1, since the logit link's inverse transformation asymptotically approaches but never exceeds 1.

#416 · Generalized Linear Models

What happens to a fitted probability from a logistic regression as the linear predictor approaches very large negative values?

Answer

The fitted probability approaches 0, asymptotically, never going below it.

#417 · Generalized Linear Models

Why is it misleading to describe an odds ratio of 1.6667 as a 66.67 percent increase in probability?

Answer

An odds ratio describes a change in odds, not probability directly, and the actual probability change depends on the baseline probability level, which is generally smaller than the odds change suggests.

#418 · Generalized Linear Models

What is one clear sign that a lasso model has performed variable selection compared to the full model?

Answer

Some coefficients in the lasso output are exactly zero, indicating those variables were effectively dropped from the model.

#419 · Generalized Linear Models

How would a candidate defend using elastic net over lasso when the exhibit shows several highly correlated territory-level variables?

Answer

Elastic net tends to retain or shrink correlated groups of variables together instead of arbitrarily keeping only one and dropping the rest, which better represents the shared information across correlated territories.

#420 · Generalized Linear Models

What should a candidate check first when an exam exhibit shows two candidate GLMs with different response distributions for the same business problem?

Answer

Whether the response variable's actual shape, such as counts, positive continuous values, or a point mass at zero, matches the assumptions of each candidate distribution.

#421 · Generalized Linear Models

Why is Poisson regression with a log link considered the natural starting point for auto claim frequency modeling?

Answer

Claim counts are non-negative integers, and the log link ensures fitted frequencies stay positive while allowing multiplicative rating relativities.

#422 · Generalized Linear Models

What warning sign in a coefficient table might suggest multicollinearity among predictors?

Answer

Unusually large standard errors or coefficients with unexpected signs that flip when a similar, correlated variable is added or removed from the model.

#423 · Generalized Linear Models

How does a candidate connect a low AIC value back to a business recommendation in a written response?

Answer

By stating that the model with the lowest AIC balances predictive accuracy against complexity, and recommending it as the model expected to generalize best to new policies.

#424 · Generalized Linear Models

Why should exam candidates verify their arithmetic when exponentiating coefficients under time pressure?

Answer

A small arithmetic slip in the exponentiation step produces a materially wrong relativity, which can undermine an otherwise sound written explanation of the model.

#425 · Generalized Linear Models

What single theme ties together response distribution choice, offsets versus weights, coefficient interpretation, and model validation on Exam PA?

Answer

Translating a fitted GLM into decisions and explanations a real insurance business can act on, always grounded in correct statistical reasoning.

#426 · Tree-Based Models and Ensemble Methods

What is a decision tree?

Answer

A predictive model that splits data into groups using a sequence of yes/no questions about the predictors, giving every observation in the same terminal leaf the same prediction.

#427 · Tree-Based Models and Ensemble Methods

What is a terminal node (leaf)?

Answer

A node at the end of a branch of a tree that is not split further. All observations reaching that leaf get the same predicted value.

#428 · Tree-Based Models and Ensemble Methods

How is the prediction in a regression tree leaf calculated?

Answer

It is the average of the target variable across all training observations that fall into that leaf.

#429 · Tree-Based Models and Ensemble Methods

How is the prediction in a classification tree leaf calculated?

Answer

It is the most common class among training observations in that leaf, or the class proportions used as estimated probabilities.

#430 · Tree-Based Models and Ensemble Methods

What is recursive binary splitting?

Answer

The greedy algorithm used to grow a tree, which at each step picks the single predictor and split point that most improves the fit, without looking ahead to future splits.

#431 · Tree-Based Models and Ensemble Methods

Why is tree-growing called a greedy algorithm?

Answer

Because it always chooses the best split available at the current step, without considering whether a different choice now might lead to a better overall tree later.

#432 · Tree-Based Models and Ensemble Methods

What criterion does a regression tree usually minimize when choosing a split?

Answer

The residual sum of squares, summed across the two branches created by the split.

#433 · Tree-Based Models and Ensemble Methods

Name the three common splitting criteria for classification trees.

Answer

Classification error rate, the Gini index, and entropy (cross-entropy).

#434 · Tree-Based Models and Ensemble Methods

What is the Gini index?

Answer

A measure of node impurity equal to the sum over all classes of the class proportion times one minus that proportion; it is small for a pure node and large for a mixed node.

#435 · Tree-Based Models and Ensemble Methods

What is entropy in the context of classification trees?

Answer

A measure of node impurity equal to the negative sum over all classes of the class proportion times the log of that proportion, behaving numerically similarly to the Gini index.

#436 · Tree-Based Models and Ensemble Methods

Why are Gini index and entropy preferred over classification error rate for growing a tree?

Answer

They are more sensitive to node purity and produce better splits during tree construction, while classification error rate is better used to evaluate a finished tree's accuracy.

#437 · Tree-Based Models and Ensemble Methods

What happens if a tree is grown without any limit?

Answer

It overfits the training data, splitting until leaves are very small and the tree has essentially memorized noise rather than signal, which generalizes poorly to new data.

#438 · Tree-Based Models and Ensemble Methods

What is cost-complexity pruning?

Answer

A technique that grows a large tree first, then prunes it back by penalizing the number of terminal nodes, using a tuning parameter often called alpha.

#439 · Tree-Based Models and Ensemble Methods

What does alpha equal to zero produce in cost-complexity pruning?

Answer

The full, unpruned tree, since there is no penalty applied for having additional terminal nodes.

#440 · Tree-Based Models and Ensemble Methods

How is the optimal value of alpha chosen?

Answer

Using cross-validation: for each candidate alpha, the tree is grown and pruned on training folds and tested on a held-out fold, and the alpha with the lowest average cross-validated error is selected.

#441 · Tree-Based Models and Ensemble Methods

Why is a lower training error alone not evidence of a better tree?

Answer

A deeper, more complex tree will almost always fit the training data better, but that improvement can reflect overfitting rather than genuinely better predictive ability on new data.

#442 · Tree-Based Models and Ensemble Methods

Give an actuarial example of a classification tree.

Answer

An auto insurer flagging potentially fraudulent claims by splitting on indicators such as time between policy inception and claim filing, prior claims history, and reported loss relative to vehicle value.

#443 · Tree-Based Models and Ensemble Methods

What is a strength of single decision trees regarding predictor types?

Answer

Trees handle both numeric and categorical predictors directly, without requiring dummy variable encoding.

#444 · Tree-Based Models and Ensemble Methods

What is the main weakness of a single decision tree?

Answer

High variance: a small change in the training data can produce a very different tree structure, and single trees generally predict less accurately than ensemble methods.

#445 · Tree-Based Models and Ensemble Methods

What is bagging?

Answer

Bootstrap aggregating: a technique that grows many full, unpruned trees on separate bootstrap samples of the training data and averages (or votes on) their predictions to reduce variance.

#446 · Tree-Based Models and Ensemble Methods

How is a bootstrap sample generated?

Answer

By randomly drawing observations from the training data with replacement, typically producing a resample the same size as the original data set.

#447 · Tree-Based Models and Ensemble Methods

How is the final prediction formed in bagging for a regression problem?

Answer

By averaging the predictions from all the individual trees grown on the different bootstrap samples.

#448 · Tree-Based Models and Ensemble Methods

How is the final prediction formed in bagging for a classification problem?

Answer

Usually by majority vote across all the trees, though averaging predicted class probabilities is also used.

#449 · Tree-Based Models and Ensemble Methods

Why are individual trees in bagging left unpruned?

Answer

Because averaging many high-variance, low-bias trees reduces variance directly, so pruning each individual tree is no longer necessary.

#450 · Tree-Based Models and Ensemble Methods

Does adding more trees to a bagged ensemble cause overfitting?

Answer

No. Unlike a single tree, adding more trees to a bagged ensemble does not itself cause overfitting; it only improves the variance-reduction benefit.

#451 · Tree-Based Models and Ensemble Methods

What is an out-of-bag (OOB) observation?

Answer

An observation from the original training data that was not selected into a given tree's bootstrap sample, and so was left out of that particular tree's training.

#452 · Tree-Based Models and Ensemble Methods

Roughly what fraction of observations are out-of-bag for a given bootstrap sample?

Answer

About one-third, since on average only about two-thirds of the original observations appear in any single bootstrap sample of the same size.

#453 · Tree-Based Models and Ensemble Methods

What is the OOB error estimate used for?

Answer

It estimates a bagged model's out-of-sample (test) error by comparing each observation's predictions, made only from the trees for which it was out-of-bag, to its actual value, without needing a separate validation set.

#454 · Tree-Based Models and Ensemble Methods

What is the key difference between a random forest and plain bagging?

Answer

At each split, a random forest considers only a random subset of predictors rather than all available predictors.

#455 · Tree-Based Models and Ensemble Methods

What is mtry?

Answer

The tuning parameter specifying how many predictors are randomly considered at each split in a random forest.

#456 · Tree-Based Models and Ensemble Methods

What is a common default value for mtry in classification problems?

Answer

The square root of the total number of predictors.

#457 · Tree-Based Models and Ensemble Methods

What is a common default value for `mtry` in regression problems?

Answer

One-third of the total number of predictors.

#458 · Tree-Based Models and Ensemble Methods

Why does restricting predictors at each split improve on plain bagging?

Answer

It prevents a single dominant predictor from being chosen for the top split of nearly every tree, which decorrelates the trees so their averaged predictions achieve a greater reduction in variance.

#459 · Tree-Based Models and Ensemble Methods

What happens to bagged trees if one predictor is very strong?

Answer

Most or all of the bagged trees tend to use that predictor for their top split, making the trees highly correlated with each other and reducing the benefit of averaging.

#460 · Tree-Based Models and Ensemble Methods

Why does averaging correlated trees reduce variance less than averaging independent trees?

Answer

Correlated errors do not cancel out as effectively as independent errors when predictions are averaged together.

#461 · Tree-Based Models and Ensemble Methods

Name two ways to measure variable importance in a random forest.

Answer

Total reduction in the split criterion (RSS or Gini index) attributable to a predictor across all trees, and the increase in OOB error when a predictor's values are randomly permuted.

#462 · Tree-Based Models and Ensemble Methods

How does the permutation-based variable importance measure work?

Answer

The values of one predictor are randomly shuffled in the OOB data, and the resulting increase in OOB error is recorded; a larger increase indicates a more important predictor.

#463 · Tree-Based Models and Ensemble Methods

Give an actuarial example of a random forest application.

Answer

Predicting policyholder churn (lapse), using variable importance to identify tenure, premium change at renewal, and prior claims count as the top drivers.

#464 · Tree-Based Models and Ensemble Methods

What is boosting?

Answer

An ensemble method that builds trees sequentially, where each new tree is specifically fit to correct the errors made by the trees already built.

#465 · Tree-Based Models and Ensemble Methods

How does boosting differ structurally from bagging?

Answer

Bagging builds trees independently and in parallel on separate bootstrap samples; boosting builds trees sequentially, each depending on the residuals left by the prior trees.

#466 · Tree-Based Models and Ensemble Methods

In gradient boosting, what is each new tree fit to?

Answer

The current residuals: what remains after subtracting the current model's predictions from the actual target values, rather than the original target itself.

#467 · Tree-Based Models and Ensemble Methods

What is the shrinkage parameter (learning rate) in boosting?

Answer

A small positive number, often between 0.001 and 0.1, that scales down each new tree's contribution before it is added to the running model.

#468 · Tree-Based Models and Ensemble Methods

What tradeoff does the shrinkage parameter create?

Answer

Smaller shrinkage values usually require more trees to reach the same fit, but the slower learning process often produces a better final model.

#469 · Tree-Based Models and Ensemble Methods

What is interaction depth in boosting?

Answer

The number of splits allowed in each individual tree added during the boosting process.

#470 · Tree-Based Models and Ensemble Methods

What is a stump?

Answer

A tree with a single split, often used as the individual tree type in boosting, since flexibility is built up through many sequential small trees rather than one complex tree.

#471 · Tree-Based Models and Ensemble Methods

Why can boosting overfit if given too many trees?

Answer

Each additional tree keeps fitting ever smaller residuals, some of which represent noise rather than real signal, so too many trees causes the model to fit noise.

#472 · Tree-Based Models and Ensemble Methods

How is the number of trees in boosting typically chosen?

Answer

Using cross-validation, unlike bagging and random forests where more trees generally does not hurt.

#473 · Tree-Based Models and Ensemble Methods

List the three main tuning parameters of gradient boosting.

Answer

Number of trees, shrinkage (learning rate), and interaction depth.

#474 · Tree-Based Models and Ensemble Methods

Does bagging reduce bias or variance?

Answer

Bagging (and random forests) primarily reduce variance by averaging many high-variance, low-bias trees.

#475 · Tree-Based Models and Ensemble Methods

Does boosting reduce bias or variance?

Answer

Boosting primarily reduces bias, building up predictive power gradually from many small, low-variance trees.

#476 · Tree-Based Models and Ensemble Methods

Is bagging naturally resistant to overfitting as trees are added?

Answer

Yes, since averaging more independent estimates only helps and does not add overfitting risk the way boosting can.

#477 · Tree-Based Models and Ensemble Methods

Does boosting use random predictor subsets like random forests?

Answer

No. Boosting controls complexity through shrinkage, interaction depth, and the number of trees, not through randomly restricting predictors at each split.

#478 · Tree-Based Models and Ensemble Methods

Which ensemble method can be fit with trees built in parallel: bagging/random forests or boosting?

Answer

Bagging and random forests, since each tree is grown independently on its own bootstrap sample.

#479 · Tree-Based Models and Ensemble Methods

Give an actuarial example comparing boosting and random forests.

Answer

A life insurer predicting lapse propensity for a term life block might find a carefully tuned boosted model outperforms a random forest, at the cost of a more involved tuning process.

#480 · Tree-Based Models and Ensemble Methods

What is the main tradeoff of using ensemble methods instead of a single tree?

Answer

Ensembles generally predict more accurately, but sacrifice the direct visual interpretability of a single tree that can be drawn and explained split by split.

#481 · Tree-Based Models and Ensemble Methods

How can interpretability be partly recovered for an ensemble model?

Answer

By reporting variable importance measures, which identify which predictors most influence the ensemble's predictions even though no single tree can be drawn.

#482 · Tree-Based Models and Ensemble Methods

What is overfitting?

Answer

When a model fits the training data very closely, including its noise, but performs poorly when applied to new, unseen data.

#483 · Tree-Based Models and Ensemble Methods

Why are trees considered scale-invariant to predictors?

Answer

Because splits are based on threshold comparisons of predictor values, so trees are not affected by whether a predictor is measured in different units or scales.

#484 · Tree-Based Models and Ensemble Methods

Do trees naturally capture interactions between predictors?

Answer

Yes, because a later split can be made conditional on the outcome of an earlier split, effectively modeling how predictors interact without specifying interaction terms manually.

#485 · Tree-Based Models and Ensemble Methods

What is a common range for the number of trees used in bagging or random forests?

Answer

Often a few hundred trees, since additional trees continue to reduce variance without introducing new overfitting risk.

#486 · Tree-Based Models and Ensemble Methods

Why is OOB error useful in a time-limited exam setting?

Answer

It gives an estimate of test error using only the training data itself, avoiding the need to set aside and re-fit on a separate holdout partition.

#487 · Tree-Based Models and Ensemble Methods

What does it mean for a criterion to measure node purity?

Answer

It quantifies how mixed the classes are within a node; a pure node contains observations almost entirely from one class, while a mixed node has classes evenly represented.

#488 · Tree-Based Models and Ensemble Methods

Why might classification error rate be a poor choice for growing a tree?

Answer

It is less sensitive to changes in node composition than the Gini index or entropy, so it can fail to distinguish between splits that actually differ meaningfully in purity.

#489 · Tree-Based Models and Ensemble Methods

What does cross-validation estimate when tuning tree size?

Answer

The expected prediction error of a tree of a given size on data it was not trained on, used to select the pruning parameter that minimizes this estimated error.

#490 · Tree-Based Models and Ensemble Methods

What role does the number of terminal nodes play in cost-complexity pruning?

Answer

It appears as a penalty term multiplied by alpha; more terminal nodes increase the penalty, discouraging overly complex trees as alpha grows.

#491 · Tree-Based Models and Ensemble Methods

What does weakest link pruning refer to?

Answer

Another name for cost-complexity pruning, describing how the algorithm sequentially removes the split that adds the least to the tree's fit relative to its added complexity.

#492 · Tree-Based Models and Ensemble Methods

Why is a single pruned tree often preferred for regulatory or stakeholder communication?

Answer

It can be drawn out and its splits explained directly in plain language, unlike an ensemble of many trees whose combined logic cannot be visualized as one diagram.

#493 · Tree-Based Models and Ensemble Methods

What should a written Exam PA response state when proposing a tree-based model?

Answer

The specific method (single pruned tree, bagging, random forest, or boosting), a justification tied to the business problem, and how tuning parameters were selected.

#494 · Tree-Based Models and Ensemble Methods

How does a random forest reduce correlation among its trees compared to bagging?

Answer

By restricting each split to a randomly chosen subset of predictors, so different trees are forced to rely on different predictors rather than all converging on the same dominant one.

#495 · Tree-Based Models and Ensemble Methods

Why is boosting generally harder to tune well than a random forest?

Answer

Boosting has three interacting tuning parameters (number of trees, shrinkage, interaction depth) that must be chosen together, while a random forest performs reasonably well with comparatively little tuning.

#496 · Tree-Based Models and Ensemble Methods

What is the effect of interaction depth being very small in boosting?

Answer

Each individual tree captures only limited structure (often just one split), and the ensemble's overall flexibility comes from combining many such simple trees sequentially.

#497 · Tree-Based Models and Ensemble Methods

Why might variable importance rankings differ between a random forest and a boosted model on the same data?

Answer

Because the two methods build trees differently (parallel decorrelated trees versus sequential residual-fitting trees), so predictors can contribute to overall accuracy in different proportions across the two approaches.

#498 · Tree-Based Models and Ensemble Methods

What is the closing structure recommended for a written tree-based model response?

Answer

State the modeling goal, propose a specific method with justification, describe how complexity was controlled, report a performance measure, and interpret the result in plain business language.

#499 · Tree-Based Models and Ensemble Methods

Why is greedy splitting a limitation rather than a fatal flaw?

Answer

It makes tree construction fast and computationally tractable, since every possible split does not need to be re-evaluated against all future splits, which matters far more than the accuracy cost of occasionally missing a globally better split sequence. In practice the resulting trees still perform well, especially once combined with pruning to control overfitting or ensembling to reduce variance, even though neither of those techniques directly corrects the greedy split search itself.

#500 · Tree-Based Models and Ensemble Methods

Summarize the core contrast between bagging, random forests, and boosting in one idea each.

Answer

Bagging averages trees from bootstrap samples to cut variance; random forests add random predictor subsets to further decorrelate those trees; boosting builds trees sequentially on residuals to gradually cut bias.